

Searching for Trading Edges Across Synthetic and Public Markets: A Reproducibility Audit and Frozen Cross-Asset Study

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Abstract

Objective. This study asks whether a large, documented programme of trading-rule research established a causal, reproducible, and economically implementable market edge, or whether the observed price changes were better characterized as random-walk or iid-like at the tested horizons.

Methods. We audited 60 experiments across 13 research roots involving synthetic indices, equities, foreign exchange, metal proxies, and cryptocurrencies. The archive included classical indicators, machine learning, evolutionary search, hidden-Markov filters, neural networks, reinforcement learning, transfer tests, prospective observation, and product-specific strategies. We then conducted a frozen daily cross-asset study of five standalone variants: two long/flat time-series trend rules, a 13-pair FX trend replication, a cross-sectional stock-momentum diagnostic, and a two-pair yield-differential carry proxy. Four embargoed walk-forward development folds were evaluated after stated one-way costs. Inference used a stationary block bootstrap with 5,000 replications and Benjamini–Hochberg correction. Advancement required all eight preregistered statistical, economic, robustness, concentration, and drawdown gates.

Results. No validation-eligible variant passed every development gate. On Volatility 75 M1, 39,999 log returns were near Gaussian (skewness -0.0079 ; excess kurtosis -0.0036); lag-one return autocorrelation was -0.0026 , Ljung–Box and variance-ratio tests did not reject, the runs-test p-value was 0.838, direction entropy was 0.9999998 bits, and walk-forward directional accuracy was 49.88% (approximate 95% interval 49.10–50.67%; permutation $p=0.58$). M5 and M15 returns showed similarly small lag-one correlations and near-Brownian Hurst estimates. Nonlinear, spectral, state-space, deep-learning, and cross-index tests detected some distributional and state structure, but none produced stable direction or a cost-surviving rule. In the frozen cross-asset study, positive development paths failed concentration or data-validity requirements; direct FX replication was negative after costs.

Conclusion. At the sampled resolutions, the dominant empirical fingerprint was random-walk or iid-compatible: directional uncertainty was close to a fair coin, serial dependence was economically negligible, and predictive models remained near chance. The evidence does not establish a reproducible, economically meaningful out-of-sample trading edge after costs. The data were not perfectly featureless—small nonlinear, tail, volatility, and regime effects appeared—but those deviations did not become validated predictability. This is a finite conclusion about the tested instruments, dates, transformations, and costs, not a proof that every market is always a random walk. Because no eligible strategy advanced, the exact one-year cross-asset holdout remained unscored.

Keywords: algorithmic trading; random walk; iid returns; market efficiency; backtest overfitting; data snooping; synthetic indices; reproducibility

JEL codes: C12, C22, C52, G11, G12, G14, G17

1 Introduction

Historical trading simulations can produce visually compelling equity curves even when no stable predictive relation exists. The problem is acute when researchers inspect many indicators, assets, horizons, models, state filters, and execution choices and retain the most profitable result. Reuse of a finite price history changes the interpretation of ordinary test statistics and can make the best observed strategy a selection artifact (Lo and MacKinlay, 1990; White, 2000). The large number of published return predictors creates the same inferential problem at field scale (Harvey, Liu, and Zhu, 2016), while strategy selection creates a specific risk of backtest overfitting (Bailey et al., 2017).

This paper studies that problem in an unusually complete local research archive. Appendix B reports every registered study individually, and Appendix A records the equations actually used. The programme began with the practical objective of finding a strategy capable of more than 300% annual net return while limiting peak-to-trough account-equity drawdown to 50%. It expanded across Deriv synthetic indices and later across stocks, currencies, metal exchange-traded funds, and spot cryptocurrency indices. The archive contains failed rules as well as selected positive results, code, ledgers, prospective observations, and successive corrections. This permits two related analyses: a retrospective reproducibility audit of the full search and a new frozen cross-asset experiment conducted after the earlier work was mapped.

The contribution is methodological and empirical. First, we distinguish reproducibility from validation: an exact replay can establish what a program computed without showing that the rule has positive expected net returns. Second, we force all candidates through a common sequence of causal execution, cost, uncertainty, multiplicity, concentration, and drawdown checks. Third, we preserve a one-year holdout when development evidence is insufficient, avoiding the common practice of consuming a test sample merely because it is available. Fourth, we state the mathematical limits of the conclusion. A finite collection of negative or inconclusive tests cannot prove universal market efficiency; a conditional martingale result applies only when its assumptions hold.

The central result is a disciplined non-discovery. Two cross-asset trend paths appear statistically and economically positive in development, yet both depend too heavily on Bitcoin to support a claim of a broad mechanism. The stock-momentum path also appears positive but is contaminated by construction of the universe. The direct FX trend replication is negative, and the carry proxy is imprecise and concentrated. None qualifies for holdout evaluation. The wider archive tells the same general story: spectacular selected results are explained by overlapping selection data, noncausal state labels, incomplete cost models, short samples, or failure to transfer.

2 Related literature and hypotheses

Time-series momentum has been documented across liquid futures in equities, currencies, commodities, and bonds (Moskowitz, Ooi, and Pedersen, 2012). Cross-sectional stock momentum has a long empirical

history (Jegadeesh and Titman, 1993), and currency momentum has been reported separately from conventional carry strategies (Menkhoff et al., 2012). Currency carry returns have also been interpreted through common risk-factor exposure rather than as mechanical arbitrage (Lustig, Roussanov, and Verdelhan, 2011). These findings motivate the economic families examined here, but they do not validate the exact data, implementation, universe, costs, or dates in this study.

The statistical design follows the literature on dependent resampling and multiple comparisons. The stationary bootstrap resamples variable-length blocks to preserve weak serial dependence (Politis and Romano, 1994). Benjamini and Hochberg (1995) provide a false-discovery-rate procedure for a declared family of hypotheses. White (2000) and Hansen (2005) address superior predictive ability after specification search. We use a simpler preregistered gate structure because the cross-asset family contains only five standalone variants and because the wider archive lacks a recoverable count of every informal trial.

The hypotheses are:

- **H1:** Long/flat trend based on positive 60- or 120-calendar-day return produces positive net returns across SPY, GLD, SLV, BTC-USD, and ETH-USD without dependence on one instrument.
- **H1-REP:** The 60-observation trend rule transfers without refitting to thirteen FX pairs previously studied elsewhere in the archive.
- **H2:** A long-only, top-five 12–1 momentum rule is positive on a twenty-name US equity panel. H2 is a diagnostic because the available panel is not point-in-time.
- **H3:** Two-session-lagged two-year yield differentials produce positive net carry-proxy returns in EURUSD and USDJPY.
- **H6:** A risk-parity combination is built only if eligible standalone components pass every development gate.

The null for each standalone test is that the mean daily net return is nonpositive. The programme-level research question is stricter: does any candidate provide independent, implementation-ready evidence of positive net expectancy and meet the annual-return and drawdown target?

3 Materials and data

3.1 Retrospective archive

The audit covers 60 registered experiments in 13 research roots. The instruments include Volatility 10/25/50/75/90/100 indices and one-second variants, Drift Switch Index 30, Range Break 200, foreign-exchange pairs, cryptocurrency CFDs or spot indices, and US equity CFDs or daily equities. Methods include classical indicators, tree models, time-series cross-validation, generator-forensic statistics, evolutionary genomes, hidden-Markov regimes, neural state prediction, reinforcement learning, rule inversion, transfer without refitting, prospective observation, and isolated product-mechanism screens.

An experiment is a frozen protocol or a distinct scored configuration with a recoverable result artifact. Appendix B reproduces all 60 entries with their hypotheses, observed results, evidence classes, defects, and primary archive sources. A cell is an asset-by-horizon or policy-by-pair evaluation within an experiment. These counts are descriptive and are not independent replications. Related projects often

reuse the same price histories, features, and model families. The total number of informal trials and the historical CPU search budget are unknown.

3.2 Frozen cross-asset inputs

The operative input catalogue contains 41 files, all verified by SHA-256 before market values were parsed. Daily adjusted-close or close proxies were available for SPY, GLD, SLV, BTC-USD, ETH-USD, thirteen FX pairs, and a twenty-name US stock panel. US, German, and Japanese two-year yields supplied H3. Table 1 summarizes the main limitations.

Sleeve	Instruments	Usable frequency	Principal limitation
H1-NEW	SPY, GLD, SLV, BTC-USD, ETH-USD	Daily union calendar	Close proxies and bps costs; no executable bid/ask; crypto funding omitted
H1-REP	13 Yahoo FX pairs	Common FX sessions	Previously inspected pairs; close proxies; no broker swap
H2	20 current large-cap names including QQQ	Common stock sessions from July 2015	Survivorship bias, unadjusted closes, dividends omitted
H3	EURUSD, USDJPY, three sovereign 2Y yields	Common FX sessions	Yield differential is a proxy for broker financing
Metals	GLD and SLV	ETF sessions	ETFs substitute for metal CFD or futures roll data

All timestamps were normalized to one UTC calendar date per instrument. H1-NEW used the union calendar and forward-filled marks only after an instrument's first observation; the tradability matrix retained actual session availability. H1-REP, H2, and H3 used common session calendars. Rate observations were backward-as-of aligned to the FX calendar and then shifted by two FX sessions. Backward filling was prohibited.

4 Methods

4.1 Research integrity and versioning

The cross-asset protocol passed through five immutable version snapshots before successful scoring. Versions v1–v3 were rejected during code review because synthetic fixtures exposed timestamp duplication, partial mixed-calendar execution, artificial fold-reset losses, missing volatility scaling, faulty rate alignment, contribution attribution errors, and incomplete cost specifications. Amendment 003 introduced a self-contained operative scorer. Its first development invocation stopped on a ledger reconciliation invariant before any report was written. Amendment 004 corrected only an unallocated same-day return-cost cross-term, added a regression assertion, and froze v5 before the successful invocation. The final packaged suite passed 73 of 73 synthetic tests.

Figure 1 shows the research decision path. The correction history matters because a freeze is scientifically useful only when the frozen implementation actually executes the written estimand.

4.2 Signals and sizing

H1-NEW evaluated two long/flat trend rules. At each Friday close, instrument j received a raw signal $I[\log(P(j,t)/P(j,t-L)) > 0]$, where L was 60 or 120 calendar observations. Positive signals shared equal raw weight; otherwise the portfolio held cash. H1-REP used the same construction with $L=60$ FX observations across thirteen pairs.

H2 ranked the twenty stock-panel members by return from 252 observations before the decision to 21 observations before the decision. The five highest-ranked names received equal long weights at month-end. The rule is comparable in spirit to cross-sectional momentum, but the restricted and survivorship-biased panel prevents an external validity claim.

H3 formed signed month-end positions from the two-year yield differences Germany minus United States for EURUSD and United States minus Japan for USDJPY. Positive and negative signals shared a gross exposure of one before volatility scaling. Carry accrued by calendar days using ACT/360.

For every variant, the raw target was scaled by the ratio of 10% annualized target volatility to the trailing 60-observation realized strategy volatility, capped at one. Annualization used 365 for the mixed crypto calendar and 252 for business-session strategies. Gross exposure could not exceed 1.0.

4.3 Walk-forward design and execution

Four development folds followed training cutoffs at 31 December 2015, 2017, 2019, and 2021. Each test interval began after a 21-observation embargo and ended at 31 December 2017, 2019, 2021, or 5 December 2023. Positions were flattened with cost at each fold boundary; net returns were concatenated into one \$10,000 account without inserting artificial reset losses.

A target became eligible one observation after its decision. For portfolios mixing seven-day crypto and weekday ETFs, execution was atomic: the full vector filled at the first close on which every currently held or nonzero-target instrument was tradable. The delay stress added one further observation. The cost stress multiplied all one-way costs by 1.5. Base costs were 2 bps for SPY, 3 bps for GLD/SLV, 10 bps for BTC/ETH, 1.5 bps for FX variants, and 5 bps for the stock panel. Costs applied only to executed changes in weight.

The event ledger recorded net return, gross return, cost, carry, executed delta, weight, equity, and per-symbol net contribution. For every event, symbol contributions were required to sum to the account return, and realized gross leverage was required to remain at or below one.

4.4 Inference and advancement gates

For each variant, 5,000 stationary-bootstrap samples estimated the distribution of mean daily net return. Mean block lengths were ten observations for weekly trend and five for monthly strategies. A centered-null bootstrap produced one-sided p-values; percentile intervals supplied the 95% interval. A fixed minimum of 36 observations applied. The five standalone p-values entered one global Benjamini–Hochberg procedure, with separate within-family correction.

A nondiagnostic strategy advanced only if all eight criteria passed:

1. At least 200 executed trade events or 36 calendar months.
2. Positive mean daily net return.

3. Lower bound of the 95% block-bootstrap interval above zero.
4. Profit factor above 1.10.
5. Positive mean under both 1.5× costs and one-extra-observation delay.
6. No instrument or year contributing more than 50% of summed positive contribution.
7. Development maximum drawdown strictly better than −40%.
8. Global and family q-values below 0.10.

The exact primary holdout was fixed as 4 September 2024 through 3 September 2025. A nonoverlapping replication period began 4 September 2025. The holdout could be scored once and only for nondiagnostic variants passing every development gate. H6-VALIDATION could use only passing H1-NEW components; H6-EXPLORATORY could also use passing H1-REP or H3 components. H2 was excluded.

4.5 Mathematical scope

Consider a linear self-financing exposure in discounted units. Let $M(t)$ be the midpoint, $F(t)$ the strategy information set, $q(t)$ a bounded $F(t)$ -measurable position held over the next interval, and $C(T)$ integrable cumulative nonnegative costs. Net gain is

$$G(T) = \sum q(t)[M(t+1) - M(t)] - C(T).$$

If $M(t)$ is an integrable martingale relative to $F(t)$, $q(t)$ is admissible, and the needed optional-sampling conditions hold, then

$$E[G(T)] = -E[C(T)] \leq 0.$$

This proposition does not establish that any studied midpoint is a martingale. It states a conditional limit: bounded causal transforms and stopping rules cannot create positive expected net gain from martingale increments after nonnegative costs. It does not automatically apply to nonlinear option-like contracts, unbounded doubling, or a market with predictable conditional drift.

5 Results

5.1 Frozen cross-asset development results

Table 2 presents the complete standalone result set. No validation-eligible row passed all eight gates.

Variant	Net total	Annualized	PF	95% mean daily interval	Max DD	Global q	Decision
H1-NEW-T60	+152.31%	+12.76%	1.206	+0.00564% to +0.06418%	−32.06%	0.0117	Fail: concentration
H1-NEW-T120	+235.79%	+17.03%	1.276	+0.01521% to +0.07512%	−34.40%	0.0040	Fail: concentration
H1-REP-T60	−13.14%	−1.78%	0.943	−0.02008% to +0.00632%	−24.69%	0.8392	Fail: mean, CI, PF, stress, multiplicity
H2-CS-121	+101.42%	+9.67%	1.188	+0.01021% to	−15.27%	0.0115	Numerical

				+0.06822%			pass; diagnostic only
H3-CARRY-2	+35.18%	+3.91%	1.109	−0.00237% to +0.03407%	−11.45%	0.0537	Fail: CI and concentration

The two H1-NEW paths passed the sample, mean, interval, profit-factor, stress, drawdown, and multiplicity criteria. Their failure is substantive rather than cosmetic. BTC contributed 0.6970 of 0.9941 summed positive net contribution for T60 and 0.8233 of 1.3203 for T120. The resulting 70.12% and 62.36% shares exceeded the frozen 50% ceiling. The evidence therefore supports profitable historical exposure to a particular dominant asset during the development periods, not the proposed generalization across equities, metals, and crypto.

The direct FX replication is the clearest negative result. It generated 129 trade events and lost 13.14% net, with PF 0.943, a mean interval spanning zero, negative stress means, and global $q=0.8392$. H3 had a positive point estimate and PF just above the threshold, but the mean interval included zero. USDJPY supplied 76.54% of positive instrument contribution. Because only seventeen rebalance events occurred, H3 satisfied the sample gate through elapsed months rather than transaction count.

H2 passed every numerical gate under the stated 5 bps overlay. It cannot be interpreted as a validated anomaly because the panel was selected from firms known to be prominent at the end of the sample, and because unadjusted closes omit dividends and corporate-action-consistent total returns. The diagnostic flag was frozen before scoring and prevents advancement.

Figures 2 and 3 show the chained development paths and exact gate outcomes. With no eligible component, H6 was not built. The primary and replication holdouts remained sealed.

5.2 Synthetic-index evidence

The full archive contains no validated positive synthetic-index result after costs. The strongest apparent exception was a selected Volatility 90 strategy with a six-state hidden-Markov allowlist. A 270-trade replay reproduced the reported timestamps, sides, prices, PnL states, and ending closed balance of \$94,314.66 from \$10,000. Reproduction established computational parity, not independent validation. The genome, hidden-Markov model, and allowlist were chosen on overlapping July–August 2026 data, and the historical state cache used Viterbi smoothing, which incorporates future observations.

Reconstruction at M1 closes gave a 51.79% maximum floating-equity drawdown, compared with 41.04% on closed balance. The floating estimate already breached the 50% constraint and still omitted intrabar extremes. A later causal test over approximately fifteen days was inconclusive after an execution correction: the gated 1% risk book returned +0.78%, while its block interval included zero; the ungated book returned −3.54%. That sign change was attributable to correction of short-stop fills rather than discovery of a new signal.

Other large programmes supplied stronger negative evidence. Volatility Genome v1 produced zero retained cells among 80 symbol-by-horizon tests. A frozen best cell subsequently had 23 matured prospective trades with mean −30.24 bps. Without refitting, the same mapping failed on 19 other

volatility indices, 37 cryptocurrencies, and seven FX pairs. A broader FX research matrix retained zero of 252 cells after multiplicity correction.

Neural next-state accuracies near 90% matched a stay-in-state baseline. Real-return state-riding strategies were negative after 1 bp per side; positive neural demonstrations used injected synthetic drift. A V75 directional classifier produced 49.88% walk-forward accuracy, with an approximate 95% interval of 49.10% to 50.67%.

5.3 Product-mechanism screens

Drift Switch Index 30 supplied a plausible persistence mechanism. A rule frozen before holdout compared five- and sixty-minute completed returns and traded when the standardized difference exceeded one in absolute value. On 977 holdout trades, mean gross return was +0.42 bps, but mean net return was −5.78 bps against approximately 6.2 bps spread. Every holdout week was negative; both directions and both execution stresses lost.

Range Break 200 supplied a distinct mean-reversion hypothesis: fade entries near a sixty-minute channel boundary toward the midpoint. Its frozen holdout contained 1,616 trades. Mean net return was −0.51 bps, PF was 0.83, and fixed-quantity PnL was −\$5,461 despite approximately 70% winners. Profitable target exits of \$21,130 were outweighed by stop losses of \$31,188 and 553 channel escapes losing \$26,677. The block interval on mean daily PnL was entirely negative.

Figure 4 contrasts apparent gross advantage with cost-aware results. Figure 5 shows the closed-balance and floating drawdowns of the selected Vol90 path. Figures 6 and 7 show the DSI30 and Range Break 200 failure mechanisms.

5.4 Distributional and random-walk evidence

“Random walk” is used here as a testable local description, not a metaphysical claim. If log price follows $X_t = X_{t-1} + \mu + \varepsilon_t$ with increments ε_t that are independent, identically distributed, or at least conditionally unpredictable in the available information set, then price levels need not be stationary while returns can be. Real financial series can depart from exact iid behavior through time-varying variance, jumps, tails, calendar effects, and state persistence without offering a profitable directional forecast after spread. The archive therefore tested marginal distribution, linear and nonlinear dependence, entropy, periodicity, diffusion residuals, long memory, dynamical complexity, cross-index structure, and actual out-of-sample prediction.

The most complete forensic sample contained 40,000 observations each for Volatility 75 M1, M5, and M15, plus 30,000 raw ticks. The M1 return series contained 39,999 usable observations from 4–31 July 2026. Its mean log return was -1.045×10^{-6} , standard deviation 0.0010363, skewness −0.0079, and excess kurtosis −0.0036. Normal was the preferred AIC and BIC distribution, and a two-component Gaussian mixture was not preferred. M5 and M15 were also classified normal in the same forensic pipeline. At coarser or resampled S5, S15, and H1 horizons, skew-normal sometimes fit better, so “near Gaussian” is a result for the principal bar samples rather than a universal distribution law.

Table 3 consolidates the randomness evidence. ACF, Ljung–Box, runs, and variance-ratio results are mutually consistent with no useful linear serial dependence on M1. The sign transition imbalance from 0.5 was only 0.000776. Across M1, M5, M15, S1, S5, S15, and tick series, lag-one return autocorrelation

generally lay between -0.0026 and $+0.0210$; the smaller H1 sample produced -0.069 . Squared-return autocorrelation was also small in these samples, so strong short-horizon volatility clustering was not the main source of predictability.

Diagnostic family	Archived result	Interpretation for the sampled data
M1 marginal distribution	$n=39,999$; mean -1.045×10^{-6} ; sd 0.0010363 ; skew -0.0079 ; excess kurtosis -0.0036 ; normal wins AIC/BIC	Near-Gaussian increments at M1
Linear dependence	ACF1(r) $= -0.00263$; ACF1(r ²) $= 0.00064$; Ljung–Box no rejection	No material short-lag linear memory
Runs and variance ratio	Runs $p=0.8375$; variance-ratio family did not reject random walk	Sign ordering and aggregated variance compatible with a random walk
Direction uncertainty	P(up) $= 0.50024$; H(direction) $= 0.99999984$ bits; entropy rate $= 0.99898$	Direction was essentially a fair coin
Conditional information	Maximum entropy reduction through six directions $= 0.00102$ bit; no persistent reduction above 0.02 bit	Longer direction histories added negligible usable information
Nonlinear dependence	MI exceeded shuffled null on M1, but BDS $p=0.285$ and 0.912 ; normalized permutation entropy $= 0.99906$	A small nonlinear flag existed without a stable iid rejection or forecast
Spectrum	Normalized spectral entropy $= 0.95481$; no suspicious power-of-two period	No stable clock or exploitable dominant cycle
Diffusion residuals	GBM residual ACF1 $= -0.00263$; absolute-residual ACF1 $= 0.00036$; featurelessness score $= 0.97099$	Diffusion was a strong practical benchmark
Long memory	DFA H $= 0.4890, 0.5057, 0.4777$ on M1/M5/M15; R/S H $= 0.5370, 0.5306, 0.5237$	Close to Brownian H $= 0.5$, with no stable long memory
Directional prediction	Six-fold walk-forward accuracy $= 0.49883$; interval $0.49099-0.50668$; permutation $p=0.58$	No out-of-sample directional skill
Cross-index structure	Mean absolute M5 correlation among V10/V25/V75 $= 0.00293$; PCA shares $0.335/0.333/0.331$	No dominant common generator factor or lead-lag source

Information-theoretic results were especially direct. M1 upward probability was 0.500238 and binary direction entropy was 0.99999984 bits out of a maximum of one. Conditioning on as many as six prior directions reduced entropy by at most 0.001019 bit, and the estimated entropy rate was 0.998980 . The normalized permutation entropy was 0.999063 . These values do not say that return magnitudes were identically distributed at every time; they say that the next sign remained almost maximally uncertain under the tested histories.

The nonlinear evidence was more mixed than the linear tests. Mutual information exceeded a shuffled null at M1 lags, with estimates approximately 0.0223–0.0310, and adversarial surrogate comparisons flagged three Gaussian-mixture BIC gaps. Yet BDS did not reject iid at the tested settings ($p=0.285$ and 0.912), recurrence was sparse, and the detected mutual information did not raise the gradient-boosting classifier above chance. The full forensic run collected 255 p-values; 24 were below 0.05 and 21 survived its reported Bonferroni and BH procedures. Many represented stationarity, integrity, or distributional differences rather than conditional direction. Counting them as 21 “edges” would be a category error: statistical non-randomness is not equivalent to a tradable conditional mean.

Spectral tests found normalized entropy of 0.9548 and no suspicious periods at 8, 16, 32, 64, 128, 256, 512, or 1,024 bars. Welch’s strongest M1 period was about 2.43 bars, while separate 90,000-bar runs suggested peaks around 4.1, 7.6, and 5.2 bars on M1, M5, and M15. The peaks changed with timeframe and did not survive as a causal trading rule. High spectral entropy and unstable peak location are consistent with broadband noise rather than a stable generator clock.

Memory estimates reinforced this conclusion. The 90,000-bar DFA Hurst estimates were 0.4890, 0.5057, and 0.4777 on M1, M5, and M15; corresponding R/S estimates were 0.5370, 0.5306, and 0.5237. Shannon entropy of an eight-bin representation was 2.9385, 2.9361, and 2.9505 bits against a 2.9957-bit maximum. On the independent V10/V25/V75 M5 cross-section, residual DFA estimates were 0.5125, 0.5219, and 0.5237. Small deviations around 0.5 changed sign by estimator or timeframe and did not yield stable trend or mean-reversion profits.

Nonlinear-dynamics models did not reveal a low-dimensional deterministic machine. Largest return-series Lyapunov exponents were 0.0303, 0.0305, and 0.0324 per bar on M1/M5/M15, implying estimated divergence horizons of only 33.0, 32.8, and 30.8 bars. Correlation dimensions were 3.319, 3.244, and 3.220 and were labelled high-complexity or noise-like. Jump-diffusion diagnostics attributed about 3.0–3.2% of variance to flagged jumps; absolute-return Hawkes fits had branching ratio about 0.05, indicating weak self-excitation. These are departures from a literal Gaussian iid model, but the resulting deep models still delivered LTC direction accuracies of 49.6%, 50.2%, and 50.0%, while five-step TFT accuracies were 49.4%, 50.7%, and 50.1%.

The cross-index test also rejected the proposed shared-generator advantage. On 89,999 aligned M5 returns for V10, V25, and V75, mean absolute pairwise correlation was 0.00293; dynamic-time-warping checks had Pearson correlations from -0.0019 to $+0.0062$. PCA explained 33.54%, 33.35%, and 33.11% in its three components—effectively equal shares rather than one dominant factor. Maximum absolute lead-lag correlation was 0.007. Engle–Granger tests found no pair cointegrated at 5%. Apparent residual backtests with 100% hit rates were mechanical consequences of constructing stationary residuals and exiting at zero, not independent market forecasts.

Finally, prediction supplied the economic falsification. In the 40,000-observation forensic challenge, logistic regression, random forest, and gradient boosting achieved mean accuracies of 50.05%, 50.02%, and 49.96%. Six ordered walk-forward folds averaged 49.88% with an approximate 95% interval of 49.10–50.67%; a permutation test gave $p=0.58$. On the 90,000-bar directional suite, bar-OFI, fluid, ODE, mutual-information, EVT, Hawkes, and RL variants clustered around 50% across M1, M5, and M15. A raw-tick OFI window reached 52.88% in development, but a later costed implementation had negative mean return. Thus the data were not mathematically proven iid, but their usable directional component

was indistinguishable from chance after validation and costs. Figure 8 summarizes the memory and prediction diagnostics.

The appropriate empirical statement is: **the tested return series were predominantly random-walk or iid-compatible at the measured horizons, and the deviations we could detect were not reproducible trading edges.** A stronger assertion that all markets are “just random” would exceed the evidence because the study covered finite instruments and dates, public assets can carry risk premia, and state-dependent variance or nonlinear contracts can exist without predictable next-period direction.

5.5 Relation to the 300% target

More than 300% annual net return requires capital to exceed four times its starting value over a complete 365-day period after costs and financing. It corresponds to approximately 12.25% compounded monthly or 0.3805% compounded per calendar day. None of the frozen candidates supplies such an observed untouched year. The eligible cross-asset development annualizations range from -1.78% to $+17.03\%$; these are descriptive rates derived from concatenated fold observations, not prospective calendar-year returns. The spectacular Vol90 path spans about 34 days, was selected on overlapping data, and breached the drawdown limit. Annualizing it would be statistically and economically unjustified.

6 Discussion

6.1 Why apparently strong results failed

Four recurring mechanisms explain the gap between attractive backtests and validated evidence.

First, selection and reuse were pervasive. The same histories informed genomes, state models, filters, and later variants. Exact replay could not remove that dependence. This is the practical data-snooping problem described by Lo and MacKinlay (1990) and White (2000).

Second, causal information sets mattered. Viterbi state paths, centered transforms, and best-state allowlists can use or encode information unavailable at the trading decision. Replacing smoothed labels with filtering reduced apparent performance, although lookahead was not the only source of selection bias.

Third, small gross advantages were economically dominated by trading frictions. DSI30 is the clearest example: approximately 0.4 bps gross could not cover approximately 6.2 bps spread. The Range Break strategy's high win rate concealed a loss distribution dominated by larger stops and boundary escapes.

Fourth, a profitable aggregate can lack breadth. The new H1 results survived costs, delay, bootstrap inference, drawdown, and multiple testing, yet most positive contribution came from Bitcoin. A concentration gate prevented a single-asset episode from being described as cross-asset validation. This is also why point estimates alone are insufficient: H3's positive aggregate depended on USDJPY and retained substantial uncertainty.

6.2 Interpretation of the preserved holdout

Not scoring the holdout is a result of the design rather than missing work. Opening a holdout after every development exercise converts it into another development sample. The frozen rule permitted a single evaluation only after a complete gate pass. Since the eligible passing set was empty, preserving the dates

4 September 2024 through 3 September 2025 protects their value for a genuinely new, independently specified study.

6.3 What the evidence says about mathematical edge

The evidence rejects a broad claim that the tested data transformations, statistical models, and mathematical rules jointly establish a tradable edge. It does not establish the stronger proposition that markets contain no exploitable conditional structure. Statistical non-rejection can reflect low power, and a failed implementation need not falsify an entire economic mechanism. Conversely, a significant development mean does not establish implementation-ready profitability when the universe is biased or the profit is concentrated.

The martingale proposition provides a conditional explanation for many failures: if the strategy's observable information does not predict future midpoint increments, costs make expected net gain nonpositive. The proposition is not an empirical proof of the premise. Public markets can contain risk premia, institutional frictions, and time-varying expected returns; synthetic products may also contain disclosed conditional mechanisms. The scientific claim remains bounded to the tested specifications.

7 Limitations

The retrospective archive is heterogeneous. Samples, estimands, confidence levels, and cost models differ, so the paper does not pool all trades into one meta-analytic effect. Many experiments share histories and are not independent. The number of abandoned or informal trials is unknown, preventing a complete programme-wide adjustment such as White's Reality Check or Hansen's SPA test.

The cross-asset study uses daily proxy data. It lacks executable bid/ask histories, market impact, broker-specific financing, metal futures roll series, and crypto perpetual funding. Bps overlays do not reproduce adverse selection or nonlinear slippage. GLD and SLV are fund proxies rather than spot-metal CFDs. H3 uses sovereign yields rather than actual swap schedules.

The H2 panel is unsuitable for confirmatory inference because it is not point-in-time and lacks adjusted closes. H1-REP and H3 are researcher-exposed because related FX data were inspected in earlier projects. H1-NEW is cleaner but still a development study whose apparent profits are dominated by Bitcoin. Benjamini–Hochberg correction was prespecified, but the five strategy statistics are dependent; the gate should not be interpreted as universal false-discovery control under arbitrary dependence.

The Vol90 floating drawdown is reconstructed at M1 closes rather than every tick and may understate intrabar drawdown. Later causal evidence covers only about fifteen days. Demo fills are not real-money evidence. No result demonstrates capacity, operational reliability, tax treatment, or live execution at material scale.

8 Conclusion

Across 60 registered experiments and a new frozen cross-asset study, no strategy established a reproducible, economically meaningful, out-of-sample edge after costs. The new study found positive development paths in multi-asset trend and a biased stock diagnostic, but no nondiagnostic candidate

passed every preregistered gate. The direct FX trend replication was negative, the carry proxy remained uncertain and concentrated, and the untouched annual holdout remained sealed.

The evidence also does not support more than 300% annual net return with maximum account-equity drawdown no greater than 50%. The one path with a comparable short-period return was selected on overlapping data, used noncausal labels, covered only about 34 days, and exceeded the drawdown constraint on floating equity.

The appropriate conclusion is therefore empirical and conditional: no mathematical or statistical edge was established by the tested programme. Universal impossibility would require assumptions and evidence that these studies do not provide.

Declarations

Funding. No external funding was reported for this research programme.

Competing interests. No competing financial interest was reported.

Human participants and animals. The study used market data and software artifacts only.

Trading activity. No order was sent and no live account was changed during the audit or cross-asset study.

Data and code availability. The submission package includes the manuscript, figures, a 60-row experiment registry, a 43-claim evidence map, correction lineage, frozen cross-asset inputs by hash, operative source code, synthetic tests, development ledgers, and run manifests. Raw market files remain in their recorded local sources where redistribution rights are uncertain.

Author contributions. Author details and contribution statements should be completed before journal submission.

Ethics and consent. Not applicable.

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Appendix A. Equation catalogue and disciplinary provenance

This appendix records the mathematical objects actually implemented, calculated, or used to interpret the archived studies. It separates accounting identities from predictive hypotheses and separates diagnostic models from validated trading rules. Symbols vary slightly among original projects; the notation below is harmonized for this paper. P_t is a close or midpoint, A_t and B_t are ask and bid, $w_{\{j,t\}}$ is the portfolio weight of instrument j , E_t is equity, $c_{\{j,t\}}$ is one-way cost, and F_t is the information available at decision time t . None of these equations supplies an edge by itself. An edge would require stable predictive content and positive net performance under causal execution.

A.1 Return, execution, and account mathematics

Arithmetic and logarithmic returns (implemented throughout).

$$R_t = P_t/P_{t-1} - 1, \quad r_t = \ln(P_t/P_{t-1}). \quad (A1)$$

Log returns were used for diagnostics, trend signals, diffusion residuals, volatility, entropy discretization, and several model features. Account engines generally compounded simple net returns.

Bid/ask trade profit (implemented in tick and CFD replications). For quantity q and contract multiplier m ,

$$\Pi_{long} = qm(B_{exit} - A_{entry}), \quad \Pi_{short} = qm(B_{entry} - A_{exit}). \quad (A2)$$

This identity exposed the Vol90 short-stop defect: a short stop triggered and filled at the ask; adding the spread again charged it twice.

Portfolio return with turnover cost and carry (implemented in the frozen cross-asset scorer).

$$R^{net}_t = \sum_j w_{\{j,t-1\}} R_{\{j,t\}} + Carry_t - \sum_j c_{\{j,t\}} / \Delta w_{\{j,t\}}. \quad (A3)$$

Per-symbol contributions were required to reconcile exactly to this account return. The corrected scorer also allocated the same-day interaction between price movement and execution cost so the ledger identity held.

Equity recursion and cumulative return.

$$E_t = E_{t-1}(1 + R^{net}_t), \quad R_{\{0:T\}} = E_T/E_0 - 1. \quad (A4)$$

Drawdown and maximum drawdown.

$$DD_t = E_t / \max_{\{0 \leq s \leq t\}} E_s - 1, \quad MDD = \min_t DD_t. \quad (A5)$$

The relevant Vol90 risk finding used marked equity while positions were open. Closed-balance drawdown understated that path's risk.

Annualized growth and the stated return hurdle. For D elapsed calendar days,

$$CAGR = (E_T/E_0)^{\{365/D\}} - 1. \quad (A6)$$

The requested “more than 300% annual net” condition is $E_{t+365}/E_t > 4$ after all costs. The equivalent constant rates are $4^{1/12} - 1 \approx 12.25\%$ per month and $4^{1/365} - 1 \approx 0.3805\%$ per calendar day. The paper did not annualize short selected paths as evidence.

Profit factor and Sharpe ratio.

$$PF = \sum_{i: \Pi_i > 0} \Pi_i / |\sum_{i: \Pi_i < 0} \Pi_i|, \quad SR = \sqrt{A} \cdot \text{mean}(R_t) / \text{sd}(R_t). \quad (A7)$$

A is 252 or 365 according to the calendar. Profit factor was a frozen gate in the cross-asset study; Sharpe estimates appeared in earlier audits but were never sufficient alone.

A.2 Trading-rule equations

Time-series trend and transfer rule.

$$s_{j,t}(L) = 1\{\ln(P_{j,t}/P_{j,t-L}) > 0\}. \quad (A8)$$

L was 60 or 120 calendar observations for the SPY/GLD/SLV/BTC/ETH sleeve and 60 sessions for the 13-pair FX replication. Positive signals shared equal gross weight before volatility scaling; otherwise capital remained in cash.

Cross-sectional 12–1 stock momentum diagnostic.

$$M_{j,t}^{12-1} = P_{j,t-21}/P_{j,t-252} - 1. \quad (A9)$$

The five highest values received equal long weights. Its positive result remained ineligible because the twenty-name universe was selected with future membership knowledge and its closes omitted dividends and corporate-action-consistent total returns.

Yield-differential carry proxy.

$$d^{EURUSD}_t = y^{DE}_{2y,t-2} - y^{US}_{2y,t-2}, \quad d^{USDJPY}_t = y^{US}_{2y,t-2} - y^{JP}_{2y,t-2}. \quad (A10)$$

Signed exposure followed $\text{sign}(d_t)$, and daily proxy carry accrued as $|w_t|d_t \cdot \text{days}/360$. The two-session lag and ACT/360 convention were implemented. Sovereign yields were proxies, not actual broker swap schedules.

Target-volatility scaling.

$$\hat{\sigma}_t = \text{sd}(R_{t-59:t})\sqrt{A}, \quad k_t = \min(1, 0.10/\hat{\sigma}_t), \quad w_t = k_t w_t^{\text{raw}}. \quad (A11)$$

Gross exposure was capped at one. Missing early volatility produced no leveraged extrapolation.

Drift Switch standardized return contrast.

$$Z_t = [r_t^{(5)} - \mu_t(r^{(60)})] / \sigma_t(r^{(60)}), \quad \text{trade when } |Z_t| \geq 1. \quad (A12)$$

The frozen DSI30 implementation used completed M1 returns, ten-minute decision times, entry at the first later M1 open, and a five-minute exit. Its small gross drift was far below the spread.

Range-channel location for the Range Break screen.

$$L_t = \min(P_{t-59:t}), \quad H_t = \max(P_{t-59:t}), \quad u_t = (P_t - L_t)/(H_t - L_t). \quad (A13)$$

The fade rule treated $u_t \leq 0.10$ and $u_t \geq 0.90$ as boundary regions and targeted the channel midpoint $(L_t + H_t)/2$. A high win rate did not offset larger stops and channel escapes.

A.3 Descriptive statistics and econometrics

Sample moments and standardization.

$$\bar{x} = n^{-1} \sum_t x_t, \quad s^2 = (n-1)^{-1} \sum_t (x_t - \bar{x})^2, \quad z_t = (x_t - \bar{x})/s. \quad (A14)$$

These quantities supported feature scaling, rule thresholds, confidence procedures, and the DSI screen.

Autocorrelation and Ljung–Box statistic.

$$\hat{\rho}_k = \sum_{t=k+1}^n (x_t - \bar{x})(x_{t-k} - \bar{x}) / \sum_{t=1}^n (x_t - \bar{x})^2, \\ Q(h) = n(n+2) \sum_{k=1}^h \hat{\rho}_k^2 / (n-k). \quad (A15)$$

The synthetic-index fingerprinting used these diagnostics to test serial dependence over declared lags.

Augmented Dickey–Fuller regression.

$$\Delta x_t = \alpha + \beta t + \gamma x_{t-1} + \sum_{i=1}^p \delta_i \Delta x_{t-i} + \varepsilon_t. \quad (A16)$$

ADF and related stationarity checks characterized prices, returns, and residuals. Rejecting a unit root in a transformed series was not interpreted as a cost-covering strategy.

Hurst and detrended-fluctuation scaling.

$$F(s) \propto s^H. \quad (A17)$$

$H \approx 0.5$ was treated as diffusion-like scaling over the measured range; H alone did not identify a trading direction or execution rule.

A.4 Information theory and spectral analysis

Shannon entropy, conditional entropy, and mutual information.

$$H(X) = -\sum_x p(x) \log_2 p(x), \quad H(Y|X) = -\sum_{x,y} p(x,y) \log_2 p(y|x), \\ I(X;Y) = \sum_{x,y} p(x,y) \log_2 [p(x,y)/(p(x)p(y))]. \quad (A18)$$

These diagnostics measured uncertainty and lagged dependence in discretized directions or states. Detectable mutual information was required to exceed the economic cost threshold before it could support a trading claim.

Block entropy and entropy rate.

$$H_n = H(X_1, \dots, X_n), \quad h = \lim_{n \rightarrow \infty} (H_n - H_{n-1}). \quad (A19)$$

Finite block estimates were used to assess whether longer symbol histories reduced uncertainty beyond one-step persistence.

Discrete Fourier transform, periodogram, and spectral entropy.

$$X_k = \sum_{t=0}^{n-1} x_t \exp(-i2\pi kt/n), \quad I_k = |X_k|^2/n, \quad H_{spec} = -\sum_k p_k \log p_k, \quad p_k = I_k / \sum I_k. \quad (A20)$$

FFT, periodogram, and Welch estimates searched for stable cycles. Spectral peaks or low spectral entropy were descriptive unless a causal costed rule survived walk-forward testing.

A.5 Physics-derived stochastic and dynamical models

Brownian diffusion and geometric Brownian motion.

$$dX_t = \mu dt + \sigma dW_t, \quad dS_t = \mu S_t dt + \sigma S_t dW_t \quad (A21)$$

These null models supplied residual and simulation benchmarks. Their use was diagnostic; fitting a diffusion did not prove that quoted prices were exact Brownian motions.

Ornstein–Uhlenbeck mean reversion.

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t. \quad (A22)$$

The discrete regression $\Delta X_t = a + bX_{t-1} + \varepsilon_t$ implied an estimated half-life $-\ln(2)/b$ when $b < 0$. OU fits were screened for stable, executable mean reversion and did not establish one.

GARCH(1,1) conditional variance.

$$\varepsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2. \quad (A23)$$

Volatility clustering was modeled as conditional scale, not automatically as directional predictability.

Jump-diffusion diagnostic.

$$dS_t/S_{t-} = \mu dt + \sigma dW_t + (J-1)dN_t \quad (A24)$$

Large residual moves and tail-event families were compared with this diffusion-plus-jump structure in the advanced and Crash/Boom work. The diagnostic did not produce a validated costed rule.

Largest Lyapunov exponent and forecast horizon.

$$\|\delta x_t\| \approx \|\delta x_0\| e^{\lambda t}, \quad T_{predict} \approx 1/\lambda. \quad (A25)$$

Positive estimated λ in short embeddings implied rapid divergence; the archive reported λ around 0.03 per bar in one study, giving only a short practical horizon before costs and estimation error.

Correlation dimension.

$$C(r) = 2[N(N-1)]^{-1} \sum_{i < j} 1\{\|x_i - x_j\| < r\}, \quad D_2 = \lim_{r \rightarrow 0} d \log C(r) / d \log r. \quad (A26)$$

Noise-like scaling in reconstructed state space argued against a low-dimensional deterministic generator at the tested resolutions.

Option-like delta-gamma approximation.

$$\Delta V \approx \Delta \cdot \Delta S + \frac{1}{2} \Gamma (\Delta S)^2 + \Theta \Delta t. \quad (A27)$$

Gamma-scalping and variance-harvesting studies used this physics-adjacent curvature decomposition. Their economics depended on the assumed fee-to-implied-volatility mapping; fair-volatility and costed cases did not establish an edge.

A.6 State-space, machine-learning, and reinforcement-learning equations

Hidden-Markov filtering. For transition matrix a_{ij} and emission density $b_j(x_t)$,

$$\alpha_t(j) \propto b_j(x_t) \sum_i \alpha_{t-1}(i) a_{ij}. \quad (A28)$$

This causal filter uses observations only through t . It was the required state estimate in later Vol90 testing.

Viterbi decoding.

$$\delta_t(j) = b_j(x_t) \max_i [\delta_{t-1}(i) a_{ij}]. \quad (A29)$$

Backtracking the completed Viterbi path uses later observations to label earlier states. It was suitable for descriptive segmentation but invalid as a contemporaneous trading input; that distinction materially affected the Vol90 interpretation.

Logistic probability and cross-entropy.

$$p_t = [1 + \exp(-\beta^T x_t)]^{-1}, \quad L = -\sum_t [y_t \log p_t + (1 - y_t) \log(1 - p_t)]. \quad (A30)$$

Logistic classifiers and neural heads used probability losses of this form. Near-50% directional accuracy and state accuracy explained by persistence did not become positive net expectancy.

Softmax multiclass state model.

$$p(y=k|x) = \exp(z_k) / \sum_l \exp(z_l), \quad L = -\sum_t \log p(y_t|x_t). \quad (A31)$$

This family described next-state and destination classification. Evaluation against a stay-in-state baseline was essential because class persistence made raw accuracy misleading.

Bellman return and PPO clipped objective.

$$G_t = \sum_{k=0}^{\infty} \gamma^k r_{t+k+1}, \quad L^{\text{CLIP}}(\theta) = E[\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t)]. \quad (A32)$$

PPO appeared in the archived RL projects. Its optimized reward was shaped and sometimes omitted spread; therefore episode reward values were not interpreted as percentage return, CAGR, or validated trading profit.

A.7 Resampling, multiplicity, and decision rules

Stationary block bootstrap. With restart probability $p=1/L$, the resampled index satisfies

$$L_{t+1} = L_t + 1 \bmod n \text{ with probability } 1-p; \text{ otherwise } L_{t+1} \sim \text{Uniform}\{1, \dots, n\}. \quad (A33)$$

The cross-asset scorer used 5,000 replications, with mean block lengths $L=10$ for weekly trend and $L=5$ for monthly strategies.

Centered-null one-sided p-value.

$$\hat{p} = [1 + \sum_{b=1}^B 1\{m_b^* - \bar{m} \geq \bar{m}\}] / (B+1). \quad (A34)$$

Equivalent centered formulations were used in code; the +1 numerator and denominator correction avoided a zero Monte Carlo p-value. Percentile bootstrap quantiles formed the reported interval.

Benjamini–Hochberg false-discovery-rate adjustment. For ordered p-values $p_{(1)} \leq \dots \leq p_{(m)}$,

$$q_{(i)} = \min_{j \geq i} \{mp_{(j)} / j\}, \text{ capped at 1 and made monotone.} \quad (A35)$$

The five standalone cross-asset variants entered one global family, with separate within-family q-values. Earlier heterogeneous archive results were not retroactively pooled into this correction.

Positive-contribution concentration gate.

$$C_{inst} = \max_j P_j^+ / \sum_k P_k^+, \quad C_{year} = \max_y P_y^+ / \sum_z P_z^+, \quad P^+ = \max(P, 0). \quad (A36)$$

Both ratios had to be at most 0.50. This gate rejected both positive cross-asset trend rows and the carry proxy because one instrument supplied most of the apparent gain.

Conditional martingale result with costs.

$$G_T = \sum_t q_t (M_{t+1} - M_t) - C_T, \quad E[G_T / \mathcal{F}_0] = -E[C_T / \mathcal{F}_0] \leq 0. \quad (A37)$$

This is a conditional mathematical statement. It applies only if M_t is a martingale in the strategy information set, q_t is admissible, and costs are nonnegative. The studies did not prove the premise for every market.

A.8 Disciplinary boundary, including chemistry

The archive used mathematics, probability, statistics, econometrics, information theory, nonlinear dynamics, stochastic processes from mathematical physics, control and reinforcement-learning notation, and financial-accounting identities. No chemistry-specific kinetic, equilibrium, molecular-dynamics, Arrhenius, or reaction-diffusion equation was implemented in a scored strategy or inferential test. “Shannon’s Demon” referred to a thermodynamic and information-theory analogy for rebalancing, not a chemical model. Chemistry is therefore recorded as **not used**, rather than retrofitted into the study after results were known.

The equation catalogue is exhaustive at the family level for recoverable mathematical work. Individual code paths sometimes used algebraically equivalent normalizations, discretizations, or optimizer losses. Their scientific status is governed by the associated experiment entry in Appendix B.

Appendix B. Complete study-by-study evidence register

Appendix B places every one of the 60 registered experiments inside the paper. The registration unit is a frozen protocol or distinct scored configuration with a recoverable result artifact; closely related parameter cells remain nested within their experiment. This avoids presenting dependent symbol-by-horizon cells as independent studies. The accompanying machine-readable registry preserves the original wording and delimiters.

B.1 Coverage and completeness

The register contains **60 experiments** from **12 recorded source-root labels**. The supplementary source inventory contains 1,560 hashed artifact records. The root labels and experiment counts are:

Recorded source root	Experiments
DRL_Trader	1
Downloads/bots	1
NEURAL_NET	3
PAKANG	1
SYNTHETIC_TEST	1
Trading_FX	1
Trading_pro	1
cross_asset_edge_research	6
llm-mt5-trader	38
llm-mt5-trader / Codex 2026-08-21	1
this_task	5
trading_bot	1

Outcome classification is descriptive: inconclusive=18, invalid or biased=9, planned or implementation=2, simulation plumbing=1, unverified positive claim=2, valid negative=28. No row is classified as a validated positive result after costs on a frozen independent test.

B.2 Individual studies

EXP-001. Classical Stats — Volatility 75 Index

Hypothesis. V75 is not a random walk at M1/M5/M15; indicators/ML have directional edge

Design and observed result. M1/M5/M15 ; varies ; M1 2026-05-30 to 2026-07-31; M5 2025-09-22 to 2026-07-31; M15 2024-01-06 to 2026-07-31 ; 90000 bars x 3 ; in-sample plus walk-forward RSI/SMA ; not primary for fingerprint tests ; near-Gaussian $H \sim 0.5$ $ACF \sim 0$; ML acc 50.29/50.14/49.79 percent ; bootstrap Sharpe CI includes 0 ; no robust edge

Assessment. valid negative on these bars/horizons **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. bar not tick; costs not in fingerprint tests

Primary archive sources. 11m-mt5-trader/FINAL_REPORT.md; backtest_results/REPORT.md

EXP-002. Advanced Quant — Volatility 75 Index

Hypothesis. HMM/copula/FFT/entropy/OU/quantum S/R yield tradable structure

Design and observed result. M1/M5/M15 ; varies ; same 90k windows ; 90000 x 3 ; descriptive plus some strategy overlays ; mixed ; structure tests mostly null ; no production edge ; n/a ; no usable edge

Assessment. valid negative for tested transforms **Classification:** valid negative; **evidence class:** development result.

Defects or limits. multiple unadjusted tests

Primary archive sources. advanced_results/ADVANCED_REPORT.md

EXP-003. Chaos Tda — Volatility 75 Index

Hypothesis. low-dimensional chaos/TDA/Hawkes/TFT predict direction

Design and observed result. M1/M5/M15 ; short ; same 90k ; 90000 x 3 ; in-sample dynamics ; not always costed ; $\lambda > 0$ but horizon short; D2 noise-like ; deep models $\sim$ coin flip ; n/a ; not exploitable after costs

Assessment. valid negative for these embeddings **Classification:** valid negative; **evidence class:** development result.

Defects or limits. practical horizon vs costs not a full tick book

Primary archive sources. advanced2_results/ADVANCED2_REPORT.md

EXP-004. Sectional — V10/V25/V75

Hypothesis. V10/V25/V75 share extractable generator noise

Design and observed result. M5 ; n/a ; 2025-09-22 to 2026-07-31 aligned ; 90000 triplets ; residual retests ; n/a ; essentially independent on M5 ; no shared tradable residual ; n/a

Assessment. no cross-index noise alpha **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. M5 only

Primary archive sources. sectional_results/SECTIONAL_REPORT.md

EXP-005. Directional Micro — V75

Hypothesis. OFI/lead-lag/RL/fluid/EVT give ≥ 52 percent directional skew

Design and observed result. M1/M5/M15 plus ticks ; short ; 90k bars plus tick subset ; mixed ; walk-forward claimed ; 2 bps or mid-spread ; bar-level no ≥ 52 percent; tick OFI w1000 ~ 52.9 percent hypothesis ; costed OFI mean negative in next-frontier ; WF stable > 52 percent False ; no production directional edge

Assessment. valid negative bars; tick OFI inconclusive **Classification:** valid negative; **evidence class:** development result.

Defects or limits. tick window short; mixed windows

Primary archive sources. `directional_results/DIRECTIONAL_REPORT.md`;
`next_frontier_results/NEXT_FRONTIER_REPORT.md`

EXP-006. Non Directional — V75

Hypothesis. Shannon demon / grid / gamma / exec RL profit without direction

Design and observed result. M1/M5/M15 ; path-dependent ; 90k ; mixed ; assumed fair vol and spread ; fair-IV gamma loses; grids fail with SL+time ; Shannon excess tiny; exec $\ll 1$ bp ; n/a ; no production non-directional edge

Assessment. valid negative under stated cost **Classification:** valid negative; **evidence class:** development result.

Defects or limits. multiplier IV not live-priced

Primary archive sources. `non_directional_results/NON_DIRECTIONAL_REPORT.md`

EXP-007. Next Frontier — V75 plus Crash/Boom/Step

Hypothesis. PRNG lead-lag; multiplier IV vs RV; 1000-tick OFI+RL; crash/boom tails

Design and observed result. mixed ; mixed ; ticks and bars ; mixed ; fee-IV model RV $>$ IV 100 percent of bars in sample ; gated ret 13.26 under fee-IV vs fair fixed -1.77 ; w1000 hit 0.521 costed mean - 0.00015 ; lead-lag no; OFI not production; IV requires live fee map

Assessment. IV result is model not fill; OFI fragile **Classification:** inconclusive; **evidence class:** development result.

Defects or limits. live multiplier fee-to-IV not independently reverse-engineered here

Primary archive sources. `next_frontier_results/NEXT_FRONTIER_REPORT.md`

EXP-008. Random Entry — V75 and EURUSD

Hypothesis. asymmetric exits/sizing create operational alpha with coin-flip entry

Design and observed result. M1/M5 ; TTL/stops ; 90k each ; Full RT V75 M5 -0.168; M1 -0.558 ; naive vs full overlay ; 2.0 bps V75 / 1.0 bps EURUSD ; zero-cost V75 M5 +1.43 ; full RT still negative; V75 M5 maxDD -0.51 ; $p(\text{mean} > 0)$ not a validated edge

Assessment. risk overlays improve vs naive but do not create +EV after costs **Classification:** valid negative; **evidence class:** development result.

Defects or limits. return units are path totals not annual account percent; DD from engine equity

Primary archive sources. random_entry_results/RANDOM_ENTRY_REPORT.md

EXP-009. Downloaded Bots — V75

Hypothesis. downloaded MT5 bots and SR/MR/Fourier/breakout clones have edge

Design and observed result. M1/M5 ; strategy-specific ; 90k ; several named bots ; 2 bps RT ; some in-sample +EV ; walk-forward unstable ; MT5_BOT_2024 advanced +EV WF all+ False

Assessment. no production bot from this catalog **Classification:** valid negative; **evidence class:** development result.

Defects or limits. inferred rules may not match original EA

Primary archive sources. downloads_bots_results/DOWNLOADS_BOTS_REPORT.md ;
bot_backtest_results/BOT_STRATEGIES_REPORT.md

EXP-010. Invert Strategies — V75

Hypothesis. flipping losing rules yields the opposite edge

Design and observed result. M1/M5 ; mirrored SL/TP ; same ; table of orig vs flip ; 2 bps RT ; orig mixed ; flip usually worse and not +ret ; hypothesis false on this sample

Assessment. valid negative for naive inversion **Classification:** valid negative; **evidence class:** development result.

Defects or limits. lossy rules are not a negative of a profitable rule after costs

Primary archive sources. downloads_bots_results/INVERT_STRATEGIES_REPORT.md

EXP-011. Strategy Chain — V75

Hypothesis. regime-governed switching among cloned bots is +EV

Design and observed result. M1/M5 ; chain ; 90k ; chain reports ; 2 bps ; not a validated OOS chain ; no production recommendation in reports ; see reports

Assessment. not established **Classification:** inconclusive; **evidence class:** development result.

Defects or limits. regime labels from same history

Primary archive sources. downloads_bots_results/STRATEGY_CHAIN_REPORT.md

EXP-012. Evolution Phase12 — V75 family

Hypothesis. genetic search finds WF-stable structural edge

Design and observed result. M1 ; genome exits ; 40000-bar slices typical ; pop/gens in reports ; engine spread proxy ; selected genomes WF+ ; selected development not independent ; evolution found candidates

Assessment. invalid/biased as validation; development only **Classification:** invalid or biased; **evidence class:** selected historical result.

Defects or limits. objective shopping; overlapping windows

Primary archive sources. evolution_results/PHASE1_PHASE2_REPORT.md; EVOLUTION_REPORT.md

EXP-013. Evolution Phase3 — V75

Hypothesis. escalate to ≥ 20 percent WF-stable return or ≥ 70 percent WR

Design and observed result. M1 ; genome ; sample in report ; LLM mutation best ret -0.0006 WR 0.64 n=56 WF+ no; other ranks WF+ with ~ 12 percent ret ; engine ; did not hit 20 percent or 70 percent WR target ; best listed 0.1208 WR 0.52 n=322 WF+ yes

Assessment. target not met; remaining WF+ is selected **Classification:** invalid or biased; **evidence class:** selected historical result.

Defects or limits. return units not annual; selection on WF+

Primary archive sources. evolution_results/PHASE3_ESCALATION_REPORT.md

EXP-014. Elite Return First — V30/75/90/100

Hypothesis. two-sided balanced genomes portable across vol basket

Design and observed result. M1 ; genome 30m-class ; 40000 bars pop14 gens12 ; V90 ret 0.1404 n=350 PF 1.173 WF+; V75 0.1024 n=159 PF 1.344 ; engine ; 3/4 assets WF+ ; selected historical

Assessment. not annual; not independent **Classification:** invalid or biased; **evidence class:** selected historical result.

Defects or limits. same-window evolution; units not account-year

Primary archive sources. evolution_results/ELITE_RETURN_FIRST_REPORT.md

EXP-015. Report History — V25 1s and V75

Hypothesis. reverse-engineered live ReportHistory rules reproduce profit

Design and observed result. M1 ; inferred ; statement plus M1 ; source summary 242 trades PF 1.66 \$31722; inferred variants lost in replay ; replay costs ; statement cannot establish annual DD ; inferred rules failed to reproduce

Assessment. unverified positive claim for statement; valid negative for inferred replay **Classification:** unverified positive claim; **evidence class:** unverified claim.

Defects or limits. statement lacks capital path; rule inference incomplete

Primary archive sources. report_history_results/REPORT_HISTORY_STRATEGY.md

EXP-016. Hybrid Sr Rh — V25 1s / V75

Hypothesis. hybrid ReportHistory x SR proximity

Design and observed result. M1 ; hybrid ; same family ; see report ; engine ; not a qualifying annual path

Assessment. not established **Classification:** inconclusive; **evidence class:** development result.

Defects or limits. hybrid on selected history

Primary archive sources. report_history_results/HYBRID_SR_RH_REPORT.md

EXP-017. EurUSD Suites — EURUSD

Hypothesis. same statistical suites on EURUSD

Design and observed result. M1/M5/M15 ; varies ; 90k ; suites A/B/C ; spread model ; no robust analogue of 300/50 ; FX not synthetic

Assessment. do not pool into synthetic claim **Classification:** valid negative; **evidence class:** development result.

Defects or limits. different microstructure

Primary archive sources. eurUSD_research/EURUSD_SUITES_ABC_REPORT.md

EXP-018. EurUSD Rsi Oos — EURUSD

Hypothesis. RSI mean-reversion cost-aware OOS on EURUSD M5

Design and observed result. M5 ; RSI ; OOS split in report ; see json ; cost-aware ; not a Deriv vol result

Assessment. does not support synthetic 300/50 **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. FX only

Primary archive sources. eurUSD_research/rsi_oos/EURUSD_M5_RSI_OOS_REPORT.md

EXP-019. Microstate Gated Vol90 — Volatility 90 Index

Hypothesis. six-state allowlist {1,4,5,7,12,14} on evolved Vol90 genome is tradable

Design and observed result. M1 ; 30 min range SL/TP ; 2026-07-05 to 2026-08-08 ; 270 gated / 435 baseline ; states chosen from paying mega-run on overlapping window ; bid OHLC proxy; 2 lots \$10k ; gated end \$94314.66 (+843 percent); baseline \$108933 with min -\$5564 ; closed DD 41.04 percent; floating M1-close DD 51.79 percent ; spectacular selected path

Assessment. reproducible selected history; not validated; DD breach on floating reconstruction

Classification: invalid or biased; **evidence class:** selected historical result.

Defects or limits. Viterbi lookahead; overlapping selection; not tick equity; live gate later {4; 12; 14}

Primary archive sources. results/microstate_gated/EQUITY_GATED_VS_BASELINE.md;
replay_arithmetic_audit.json

EXP-020. High Vol Microstate — V75/V90/V100

Hypothesis. high-vol microstate selection produces stable balance curves

Design and observed result. M1 ; state-conditional ; see reports ; see json summaries ; historical proxy ; positive selected curves exist

Assessment. not independent confirmation of EXP-019 **Classification:** invalid or biased; **evidence class:** selected historical result.

Defects or limits. state labels and windows overlap genome work

Primary archive sources. results/high_vol_microstate/*

EXP-021. Vg V1 — 20 volatility indices

Hypothesis. 11 causal geometry features; ridge; 20 indices x 4 horizons pass cost gates

Design and observed result. 5/15/30/60m ; horizon hold ; ~67-70 days to 2026-08-22; Vol100 1s 2026-06-13 to 2026-08-22 ; 80 cells; best 396 trades ; expanding 50/12.5 folds embargo 60 ; bar spread + 2 bps stress ; 37 cells +gross; 6 + after quote; 1 + after stress (+3.37 bps CI -5.35 to 12.09) ; 0/80 keep; BH q=1 ; NO_GO_HISTORICAL

Assessment. valid negative for this frozen screen **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. short calendar; bar proxy not tick

Primary archive sources. results/volatility_genome_v1/VOLATILITY_GENOME_V1_REPORT.md

EXP-022. Vg V2 Observer — Volatility 100 (1s) Index

Hypothesis. frozen v1-selected Vol100 1s 60m model has later executable edge

Design and observed result. 60m ; 60m ; 2026-08-22 to 2026-09-05 snapshot ; 23 matured (min 100) ; no refit ; executable quotes + 2 bps stress ; n/a ; mean -30.24 bps PF 0.51 WR 39 percent CI90 -65.0 to +4.63 ; awaiting minimum sample; not validated

Assessment. inconclusive (underpowered; point estimate negative) **Classification:** inconclusive; **evidence class:** prospective paper observation.

Defects or limits. n=23; one 37h-late outcome; STATUS mutable

Primary archive sources. results/volatility_genome_v2/STATUS.json; outcomes.json1

EXP-023. Vg V3 — Vol100 1s family

Hypothesis. fixed 4-state vol/direction overlay improves later folds

Design and observed result. 60m ; 60m ; v1 source reuse ; 129 selected vs 440 control ; +3.69 vs +4.09 bps ; historical stress ; opportunity improvement -3.01 bps CI -9.88 to 3.48 ; NO_GO

Assessment. valid negative vs control on reused history **Classification:** valid negative; **evidence class:** development result.

Defects or limits. post-hoc exploration on same history

Primary archive sources. results/volatility_genome_v3_microstates/*

EXP-024. Vg V4 — same

Hypothesis. nested causal HMM on all 11 features selects profitable states

Design and observed result. 60m ; 60m ; same source ; 207 selected +12.23 bps PF 1.31 CI -0.22 to 25.39 vs control +4.09 ; causal filter no Viterbi ; 0/160 diagnostics BH; 3/4 folds + ; strongest descriptive recent candidate fails gates

Assessment. not independent; CI includes 0 **Classification:** inconclusive; **evidence class:** development result.

Defects or limits. reused history; CI crosses 0

Primary archive sources. results/volatility_genome_v4_all_microstates/*

EXP-025. Vg V5 — Vol100 1s

Hypothesis. K=10 states 3/4/7 is a deployable demo gate

Design and observed result. 60m ; 60m ; in-sample reused ; 168 of 594 +11.02 vs -0.78 base PF 1.28 vs 0.98 ; historical stress ; activation screen PASS ; explicitly in-sample challenger

Assessment. invalid_or_biased as alpha; demo authorization only **Classification:** invalid or biased; **evidence class:** development result.

Defects or limits. no CI requirement on activation

Primary archive sources. results/volatility_genome_v5_demo_gate/*

EXP-026. Vg V6 — Vol100 1s

Hypothesis. equity-proportional sizing of v2/v5

Design and observed result. 60m ; 60m ; n/a ; sizing 100 lots per 9363.62 equity cap 300 ; n/a ; n/a ; n/a ; sizing not alpha

Assessment. not an edge test **Classification:** planned or implementation; **evidence class:** simulation plumbing.

Defects or limits. supersedes fixed-lot for demo executor

Primary archive sources.
results/volatility_genome_v6_equity_sizing/AMENDMENT_002_DEPLOYMENT_EQUITY.json

EXP-027. Vg V7 — 19 vol indices

Hypothesis. zero-refit transfer of v2+v5 to 19 other indices

Design and observed result. 60m ; 60m ; v1 histories already inspected ; 2608 pooled -9.01 bps PF 0.74 CI -12.33 to -5.77 ; spread+2bps ; 0/19 pass ; NO-GO transfer falsification

Assessment. valid negative for this transfer **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. targets previously inspected in v1; pipeline-order correction applied

Primary archive sources. results/volatility_genome_v7_cross_index_transfer/*

EXP-028. Crypto V8 — 37 crypto

Hypothesis. zero-refit vol strategy transfers to 37 crypto CFDs

Design and observed result. 60m ; 60m ; eligible 37/37 ; 11599 pooled mean -210.57 bps PF 0.05 CI -222.6 to -198.1 ; spread+2bps ; 0/37 pass ; NO-GO

Assessment. valid negative transfer (crypto) **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. wide spreads dominate; not synthetic-index claim

Primary archive sources. results/crypto_genome_v8_zero_refit_transfer/*

EXP-029. Forex V9 — 7 FX

Hypothesis. zero-refit transfer to 7 major FX

Design and observed result. 60m exact clock ; weekend/holiday excluded ; 1068 pooled stressed -3.44 bps PF 0.33 CI -3.94 to -2.94 ; spread+2bps ; 0/7 pass ; NO-GO

Assessment. valid negative transfer (FX) **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. not synthetic

Primary archive sources. results/forex_genome_v9_zero_refit_transfer/*

EXP-030. Forex V10 — 7 FX

Hypothesis. 36 frozen policies x 7 majors; any cell passes BH

Design and observed result. mixed ; mixed ; 252 cells ; 0/252 pass after BH; 22/36 +gross; 1/36 after spread; 0 after stress ; spread+2bps ; NO_GO

Assessment. valid negative matrix on previously inspected FX **Classification:** valid negative; **evidence class:** development result.

Defects or limits. prereg committed with results; data previously inspected

Primary archive sources. results/forex_genome_v10_all_research/*

EXP-031. Diversified Trend V11 — 13 FX Yahoo

Hypothesis. vol-targeted FX trend at 1-3x meets 300 percent CAGR with DD<=50 percent

Design and observed result. daily ; vol target 15 percent ; previously inspected FX ; CAGR ~0 to -1.8 percent; maxDD 27-43 percent; P(300 percent year)=0 ; Yahoo close stress ladder ; zero-cost Sharpe ~0.07 ; NO_GO; 300 percent infeasible under this rule

Assessment. valid negative for this frozen trend rule on Yahoo FX **Classification:** valid negative; **evidence class:** development result.

Defects or limits. Yahoo not executable Deriv; pairs not independent; prereg not independently timestamped

Primary archive sources. results/diversified_trend_v11/*

EXP-032. Deriv M2 — 18/19 US CFDs

Hypothesis. sealed Market Genome M2 profitable on Deriv CFD ticks

Design and observed result. signal horizon sealed ; 243 signals 25 days 2026-05-08 to 2026-08-06 ; 15 percent vol book -0.173 percent PF 0.984 after bid/ask ; zero slippage assumed ; 10 percent book PF 1.006 ; NO_GO for live; shadow only

Assessment. valid negative for 15 percent book; 10 percent economically flat **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. TXN.OQ missing; zero slippage; not synthetic vol

Primary archive sources. results/deriv_m2/DERIV_GO_NO_GO.md; Codex 2026-08-21 outputs

EXP-033. Alpaca M2 — US equities

Hypothesis. Alpaca SIP shadow of M2

Design and observed result. sealed ; telemetry ; readiness json ; n/a ; venue costs differ ; not Deriv synthetic

Assessment. do not pool **Classification:** inconclusive; **evidence class:** prospective paper observation.

Defects or limits. different venue

Primary archive sources. results/alpaca_m2/*

EXP-034. Synthetic Forensics — V75

Hypothesis. V75 generator is crackable / predictable

Design and observed result. M1/M5/M15/ticks ; n/a ; M1 40k 2026-07-04 to 07-31 ; dir acc logistic 50.05 RF 50.02 GBM 49.96; WF 49.88 CI 49.10-50.67; perm p=0.58 ; n/a ; class A iid-compatible ; no robust edge

Assessment. valid negative on sampled horizons **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. does not prove no strategy exists; uses llm-mt5-trader cache

Primary archive sources. SYNTHETIC_TEST/crack_results/CRACK_FORENSIC_REPORT.md

EXP-035. Neural Nextstate — V75/V90/V100

Hypothesis. next microstate / mapped regime accuracy ≥ 75 percent is useful for trading

Design and observed result. M1 microstates ; ~dwell ; ~30k aligned 2026-07-04/06 to 07-25/26 ; next-state 89.5-92.0 percent vs stay 89.6-91.9; full dest 47.5-52.5 percent ; 80/10/10 chrono ; n/a ; accuracy gate pass via persistence ; not directional prediction

Assessment. valid negative for destination; persistence explanation **Classification:** valid negative; **evidence class:** development result.

Defects or limits. labels from llm-mt5-trader HMM

Primary archive sources. NEURAL_NET/output/OVERNIGHT_FINAL_SUMMARY.md

EXP-036. Neural Synth Backtest — V75/V90/V100

Hypothesis. state-ride profitable on injected Gaussian state drift

Design and observed result. M1 ; dwell 3-40 ; synthetic_state_gaussian ; PnL +1.82/+1.22/+0.99 ; walk-forward 4 folds ; 1 bp/side ; plumbing demo

Assessment. NOT market evidence **Classification:** simulation plumbing; **evidence class:** simulation plumbing only.

Defects or limits. use_real_returns=false

Primary archive sources. NEURAL_NET/output/backtest/BACKTEST_SUMMARY.md

EXP-037. Neural Real Return — V75/V90/V100

Hypothesis. hybrid family/micro state-ride +EV on real M1 returns

Design and observed result. M1 ; min dwell 3 max 45 ; aligned ~12.4d test concat ; hybrid additive PnL -0.462/-0.331/-0.473; trades 2288/1563/3071 ; expanding 4 folds after 40 percent ; 1 bp/side ; all nine aggregates negative

Assessment. valid negative for this pipeline; metrics are additive not account percent **Classification:** valid negative; **evidence class:** development result.

Defects or limits. P(change) not refit inside folds; ~3 week dataset

Primary archive sources. NEURAL_NET/output/backtest/FULL_PIPELINE_REPORT.md

EXP-038. Ppo Drl Trader — V75

Hypothesis. PPO MLP on V75 M5 with Gemini labels

Design and observed result. M5 ; env episode ; 2024-09-30 to 2025-08-31 M5 ; no protocol report ; raw price delta no spread; stop at 50 percent of initial not peak DD ; saved zip exists ; no qualifying report

Assessment. unverified / incomplete **Classification:** inconclusive; **evidence class:** simulation plumbing.

Defects or limits. no spread; reward not annual return

Primary archive sources. Desktop/dev/DRL_Trader

EXP-039. Ppo Trading Bot — V75 (filename conflict with 1s possible)

Hypothesis. PPO CNN/hybrids/inverted rules on V75

Design and observed result. M15/M1 ; episode ; 2022-2025 files mixed provenance ; eval rewards last mean -0.12 to -45.6; one best 249.45 ; shaped reward ; do not treat 249 as percent

Assessment. no qualifying Deriv report **Classification:** inconclusive; **evidence class:** simulation plumbing.

Defects or limits. instrument identity of long M15 file unverified

Primary archive sources. Desktop/dev/trading_bot/logsu

EXP-040. Trading Fx Xau — XAUUSD/EURUSD

Hypothesis. classifier/ensemble on XAUUSD/FX

Design and observed result. mixed ; mixed ; performance_report_20251021 ; 3234 trades PF 1.58 maxDD -0.41 on XAUUSD ; unknown costs ; not Deriv synthetic

Assessment. do not pool into synthetic claim **Classification:** unverified positive claim; **evidence class:** unverified claim.

Defects or limits. not independently audited here

Primary archive sources. Desktop/dev/Trading_FX/reports

EXP-041. Trading Pro — mixed

Hypothesis. ensemble+PPO multi-asset including V25 1s V75 V100 1s

Design and observed result. mixed ; mixed ; logs_rl PPO ; no markdown research report found ; unknown ; no qualifying report

Assessment. missing evidence **Classification:** inconclusive; **evidence class:** unverified claim.

Defects or limits. no protocol

Primary archive sources. Desktop/dev/Trading_pro

EXP-042. Pakang V25 — Volatility 25 (1s)

Hypothesis. H1 V25 live/backtest scripts

Design and observed result. H1 ; bot ; csv historical_data/trade_log ; not a frozen scientific protocol ; unknown ; unverified

Assessment. older live-bot experiments not a 300/50 proof **Classification:** inconclusive; **evidence class:** unverified claim.

Defects or limits. credentials risk in configs; not archived as shareable

Primary archive sources. PAKANG Desktop python programs

EXP-043. Vol90 Replication Hist — Vol90

Hypothesis. reproduce 270-trade +843 percent ledger

Design and observed result. M1 ; 30m ; 2026-07-05 to 2026-08-08 ; 270/270 match end 94314.66 ; original six-state + 50k Viterbi ; historical parity engine ; reproduced

Assessment. reproduced not validated **Classification:** invalid or biased; **evidence class:** selected historical result.

Defects or limits. lookahead Viterbi; overlap

Primary archive sources. outputs/vol90_replication/04_historical_reproduction.md

EXP-044. Vol90 Causal Later — Vol90

Hypothesis. frozen genome+causal HMM +EV after 22 Aug 2026

Design and observed result. M1 ; 30m ; 2026-08-22 to 2026-09-05 ~15d ; A 78 trades -5.32 percent; B 53 -0.73 percent (pass 1) ; bid/ask 1 percent equity ; unsupported

Assessment. later edge unsupported **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. short window; then execution defects

Primary archive sources. outputs/vol90_replication/05_causal_results.md

EXP-045. Vol90 Correction 002 — Vol90

Hypothesis. same later test after SL/TP/margin/swap/gap/MTM fixes

Design and observed result. M1 ; 30m ; same later window ; A 78 -3.54 percent PF 0.89; B 53 +0.78 percent PF 1.04 CI includes 0 ; corrected engine ; B sign flip from fill convention not new edge

Assessment. still unsupported as validated edge **Classification:** inconclusive; **evidence class:** independent historical test.

Defects or limits. Amendment 003 NOT done; signed-swap and tick-spread-stress defects remain

Primary archive sources. outputs/vol90_replication/correction/REVISED_VERDICT.md; vol90_correction_pass_review.md

EXP-046. Dsi30 Z — Drift Switch Index 30

Hypothesis. Z persistence ;Z;>=1 on DSI30 survives costs

Design and observed result. M1 decisions 10m hold ~6m ; 2026-06-28 to 2026-09-05 70 days not 90 ; holdout 977 trades gross +0.42 net -5.78 bps PF 0.06 WR 14 percent ; M1 quote-proxy spread ~6.2 bps ; dev also ~ -5.55 net

Assessment. FAILED; do not transfer to DSI10/20 **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. delayed M1 proxy; 69-day coverage; one missing holdout exit

Primary archive sources. outputs/dsi_drift_screen/

EXP-047. Rb200 Fade — Range Break 200 Index

Hypothesis. fade 60-min channel 10 percent band toward mid; bounces pay for breaks

Design and observed result. M1 from ticks hold $\leq 30m$; ticks 2026-06-28 to 2026-09-05 70 days; split msc 1786830348622 ; holdout 1616 mean net -0.51 bps PF 0.83 -\$5461 WR 70 percent; daily 5-95 percent -481 to -35 USD ; tick bid/ask 1 lot; swap signed +; stresses 1.5x and +1 tick

Assessment. FAILED complete screen unresolved 0 **Classification:** valid negative; **evidence class:** independent historical test.

Defects or limits. fixed quantity not 300/50; 70 not 90 days

Primary archive sources. outputs/range_break_200_screen/

EXP-048. Vg V2 Demo Executor — Vol100 1s

Hypothesis. v2/v5/v6 demo fills are +EV

Design and observed result. 60m ; 60m ; 2026-08-22 to 2026-09-04 ; 7 completed 6 loss 1 win sum - \$2424.69 price-fill excl swap/comm ; demo fills ; negative small-n

Assessment. not a validated edge; not full account attribution **Classification:** inconclusive; **evidence class:** demo execution.

Defects or limits. n=7; account equity change not strategy PnL

Primary archive sources. results/volatility_genome_v2/demo_execution/events.jsonl

EXP-049. Scalper Genome M2 Family — US equities / mixed

Hypothesis. multiple scalper/in-play/maker/NOII/opening hypotheses

Design and observed result. intraday ; mixed ; many JSON under results/scalper_genome ; mostly no-go or sealed ; venue costs ; M2 v4 cost gate no-go in git history

Assessment. not synthetic vol 300/50 **Classification:** valid negative; **evidence class:** development result.

Defects or limits. large family; not fully re-audited trade-by-trade here

Primary archive sources. results/scalper_genome/

EXP-050. Multi Regime — V75/90/100

Hypothesis. macro/micro MoE / latent / live blueprint / trend tweak

Design and observed result. M1 ; mixed ; research_*.json ; see reports ; historical ; no standalone 300/50

Assessment. not established **Classification:** inconclusive; **evidence class:** development result.

Defects or limits. overlaps HMM/genome

Primary archive sources. results/multi_regime/*

EXP-051. Downloads Raw Bots — various

Hypothesis. dozens of MT5_BOT_* scripts are a live edge

Design and observed result. various ; bot ; source only ; no frozen OOS ledger in Downloads ; unknown ; unverified

Assessment. code without scientific freeze **Classification:** inconclusive; **evidence class:** unverified claim.

Defects or limits. possible secrets in configs excluded from shareable archive

Primary archive sources. Downloads/bots

EXP-052. Fff1 Backtest — V75

Hypothesis. fff (1).py MT5 strategy +EV

Design and observed result. M1 ; script ; 90k ; report ; engine ; no robust edge consistent with RW

Assessment. valid negative **Classification:** valid negative; **evidence class:** development result.

Defects or limits. rule extracted from script

Primary archive sources. backtest_results/FFF1_BACKTEST_REPORT.md

EXP-053. Research V2 Harness — Volatility 90 Index

Hypothesis. nested research harness on Vol90

Design and observed result. M1 ; mixed ; 2026-08-11 json ; trial_ledger ~170 trials ; historical engine ; mixed/negative outer folds

Assessment. not a KEEP **Classification:** inconclusive; **evidence class:** development result.

Defects or limits. not independently rescored here

Primary archive sources. results/research_v2/Volatility_90_Index_research_v2.json

EXP-054. Multi Regime Live Router — V75/V90/V100

Hypothesis. locked multi-regime router is a live KEEP

Design and observed result. M1 ; regime maps ; research window 80k-class ; Vol90 +38.3% Vol75 +12.3% Vol100 +6.5% routed ; not Genome v1 stress gates ; holdout Vol75 -0.59% Vol100 -12%

Assessment. selected live-window; holdouts fail; same stack as EXP-019 **Classification:** invalid or biased; **evidence class:** selected historical result.

Defects or limits. units period percent not annual account

Primary archive sources. results/multi_regime/MULTI_REGIME_REPORT.md;
IMPROVE_ROUTED_REPORT.md

EXP-055. Cross Asset Trend — SPY/GLD/SLV/BTC-USD/ETH-USD

Hypothesis. 60-calendar-day long/flat trend generalizes across equity metal and crypto proxies

Design and observed result. four embargoed walk-forward development folds ending 2023-12-05; 381 trade events; 10% vol; stated costs

Assessment. positive development path but 70.12% of positive instrument contribution from BTC; frozen concentration gate failed **Classification:** inconclusive; **evidence class:** development result.

Defects or limits. daily close proxies; no executable bid/ask; holdout sealed

Primary archive sources.

cross_asset_edge_research/runs/development_v4/SCORE_SUMMARY.json; AMENDMENT_003.md;
AMENDMENT_004.md

EXP-056. Cross Asset Trend — SPY/GLD/SLV/BTC-USD/ETH-USD

Hypothesis. 120-calendar-day long/flat trend generalizes across equity metal and crypto proxies

Design and observed result. four embargoed walk-forward development folds ending 2023-12-05; 377 trade events; 10% vol; stated costs

Assessment. positive development path but 62.36% of positive instrument contribution from BTC; frozen concentration gate failed **Classification:** inconclusive; **evidence class:** development result.

Defects or limits. daily close proxies; no executable bid/ask; holdout sealed

Primary archive sources.

cross_asset_edge_research/runs/development_v4/SCORE_SUMMARY.json; AMENDMENT_003.md;
AMENDMENT_004.md

EXP-057. Fx Trend Replication — 13 Yahoo FX pairs

Hypothesis. 60-session long/flat trend transfers across thirteen FX pairs

Design and observed result. four embargoed walk-forward development folds; 129 trade events; 1.5 bps one-way

Assessment. -13.14% net; PF 0.943; interval includes zero; stress negative **Classification:** valid negative; **evidence class:** researcher exposed replication.

Defects or limits. Yahoo daily close; pairs previously inspected; holdout sealed

Primary archive sources.

cross_asset_edge_research/runs/development_v4/SCORE_SUMMARY.json

EXP-058. Stock Momentum — 20 current US large-cap names including QQQ

Hypothesis. 12-1 top-five momentum works on a stock panel

Design and observed result. four embargoed walk-forward development folds; 86 trade events; 5 bps one-way

Assessment. mechanical gates pass but preregistered diagnostic is ineligible **Classification:** invalid or biased; **evidence class:** diagnostic development result.

Defects or limits. survivorship-biased current panel; unadjusted closes; dividends omitted

Primary archive sources.

cross_asset_edge_research/runs/development_v4/SCORE_SUMMARY.json;
HYPOTHESIS_REGISTRY.csv

EXP-059. Fx Carry Proxy — EURUSD/USDJPY with US/German/Japanese 2Y yields

Hypothesis. lagged two-year yield differentials predict EURUSD and USDJPY carry returns

Design and observed result. four embargoed walk-forward development folds; ACT/360 proxy carry; 17 trade events

Assessment. +35.18% development total but 95% mean interval includes zero and 76.54% contribution concentrated in USDJPY **Classification:** inconclusive; **evidence class:** researcher exposed development result.

Defects or limits. broker swap unavailable; yield proxy; only two pairs; holdout sealed

Primary archive sources.

cross_asset_edge_research/runs/development_v4/SCORE_SUMMARY.json

EXP-060. Cross Asset Ensemble — eligible cross-asset sleeves

Hypothesis. risk-parity combination of gate-passing sleeves yields a broad edge

Design and observed result. formula frozen before development; validation and exploratory eligibility separated

Assessment. not built because no nondiagnostic variant passed every development gate **Classification:** planned or implementation; **evidence class:** gate stopped.

Defects or limits. no scored ensemble; holdout sealed

Primary archive sources. cross_asset_edge_research/DEVELOPMENT_DECISION.json

B.3 Register-level interpretation

The complete register contains positive development paths, simulation demonstrations, incomplete prospective samples, and unverified claims, but no result that simultaneously supplies a frozen causal rule, independent data, executable costs, positive uncertainty bounds, robustness, acceptable concentration, and the requested drawdown condition. The registry therefore supports the bounded conclusion in the abstract while preserving the exact exceptions and unresolved cases needed to falsify it.

Supplementary material guide

The manuscript and machine-readable supplement both contain all 60 experiment records rather than selecting only the results discussed in the main text. `EXPERIMENT_REGISTRY.csv` classifies each experiment by family, instrument, evidence class, outcome, defects, and source.

`CLAIM_EVIDENCE_MAP.csv` maps 43 quantitative claims to fields or ledger calculations.

`SOURCE_INVENTORY.csv` contains hashed artifact records. `CORRECTION_LINEAGE.csv` records material corrections and unresolved defects. The complete cross-asset folder contains Amendments 001–004, freeze manifests v1–v5, 41 input hashes, operative code, 73 synthetic tests, development ledgers, and the decision file showing an empty passing set and unscored holdout.

The audit-edition paper retained in the parent output package provides detailed project chronology, expanded forensic notes, and additional figures. The present manuscript prioritizes the research question, methods, evidence, and limits expected in a journal submission.

Figure plates

Figures are generated from cited tables and ledgers. They are not freehand curves.

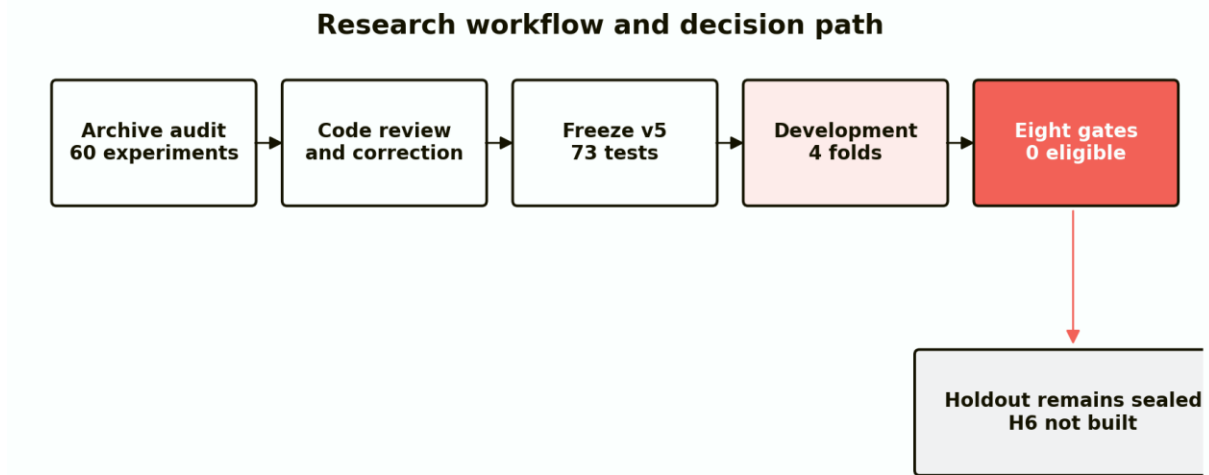


Figure 1. Audit, correction, freeze, development, and holdout decision workflow.

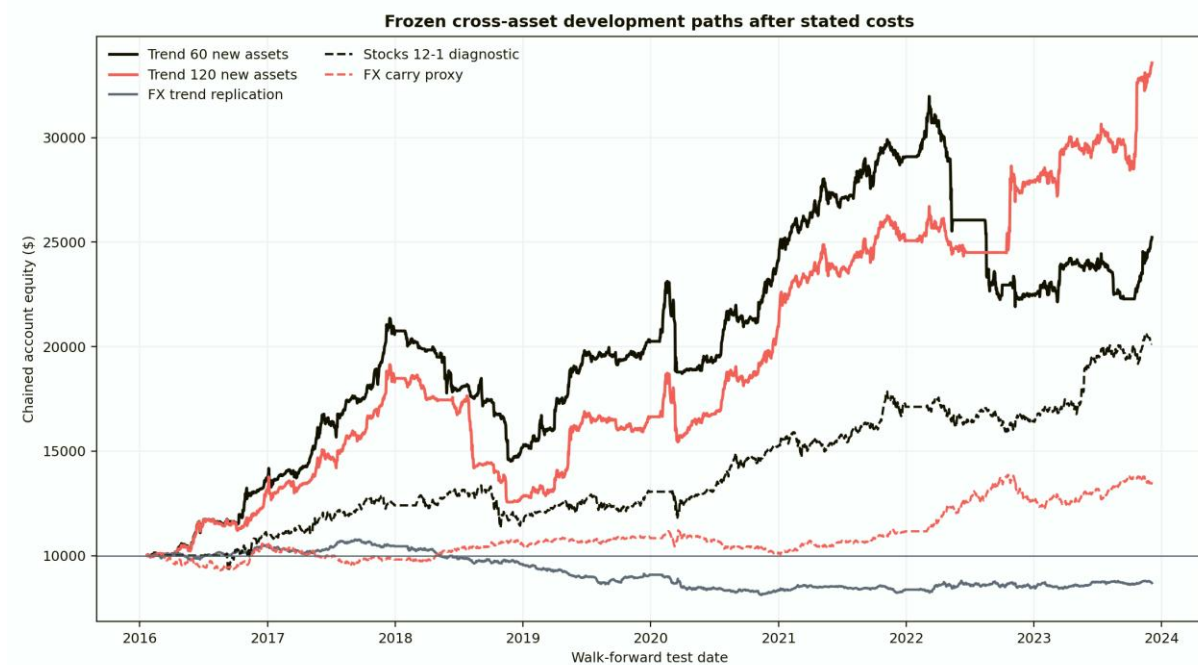


Figure 2. Frozen cross-asset development equity paths after stated one-way cost overlays.

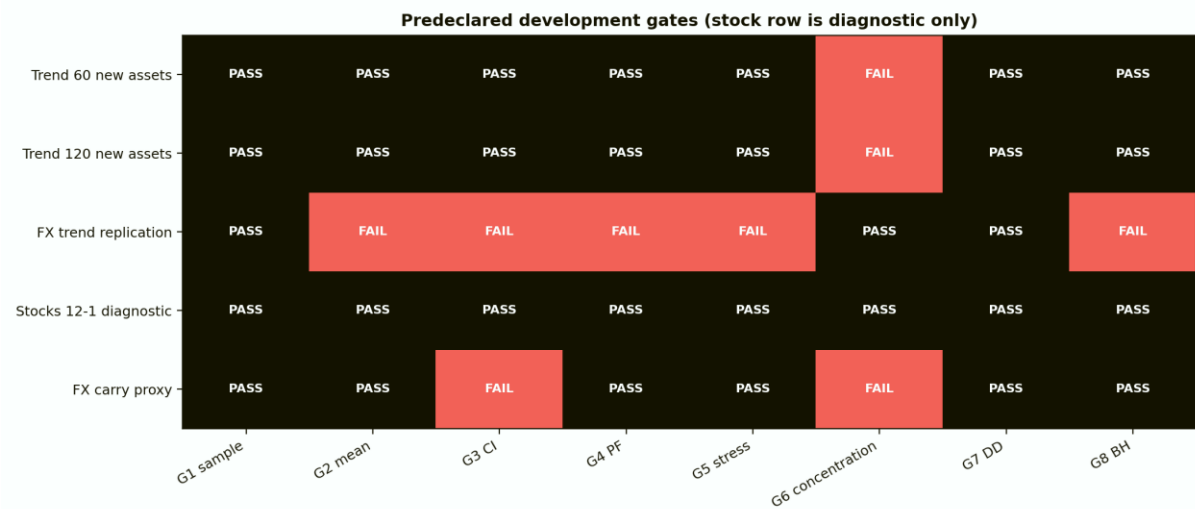


Figure 3. Predeclared development gates. H2 is diagnostic and cannot advance despite its numerical pass.

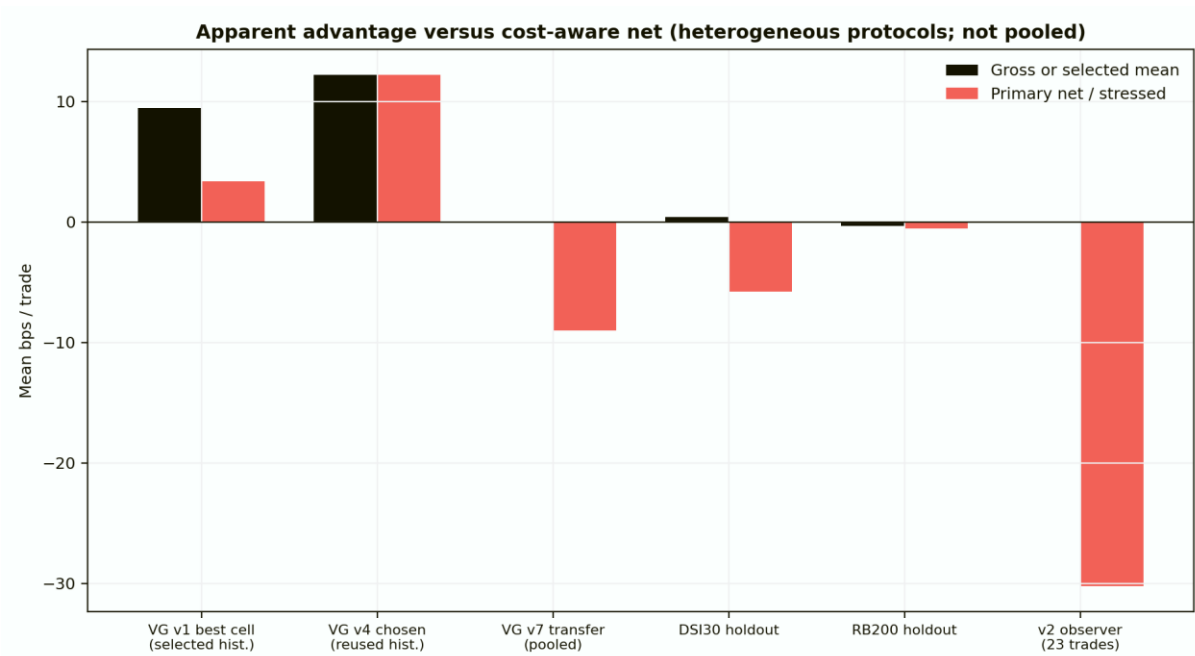


Figure 4. Selected gross advantages and cost-aware results. Protocols are heterogeneous and are not pooled.

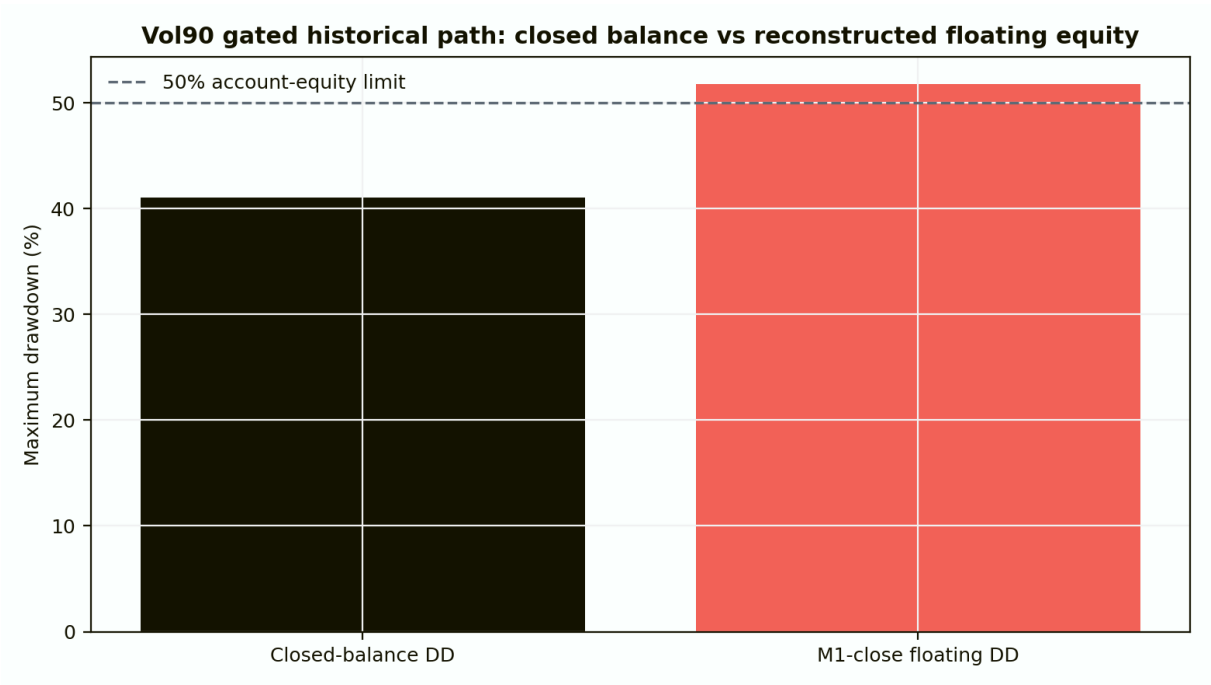


Figure 5. Selected Vol90 path: closed-balance versus reconstructed M1-close floating drawdown.

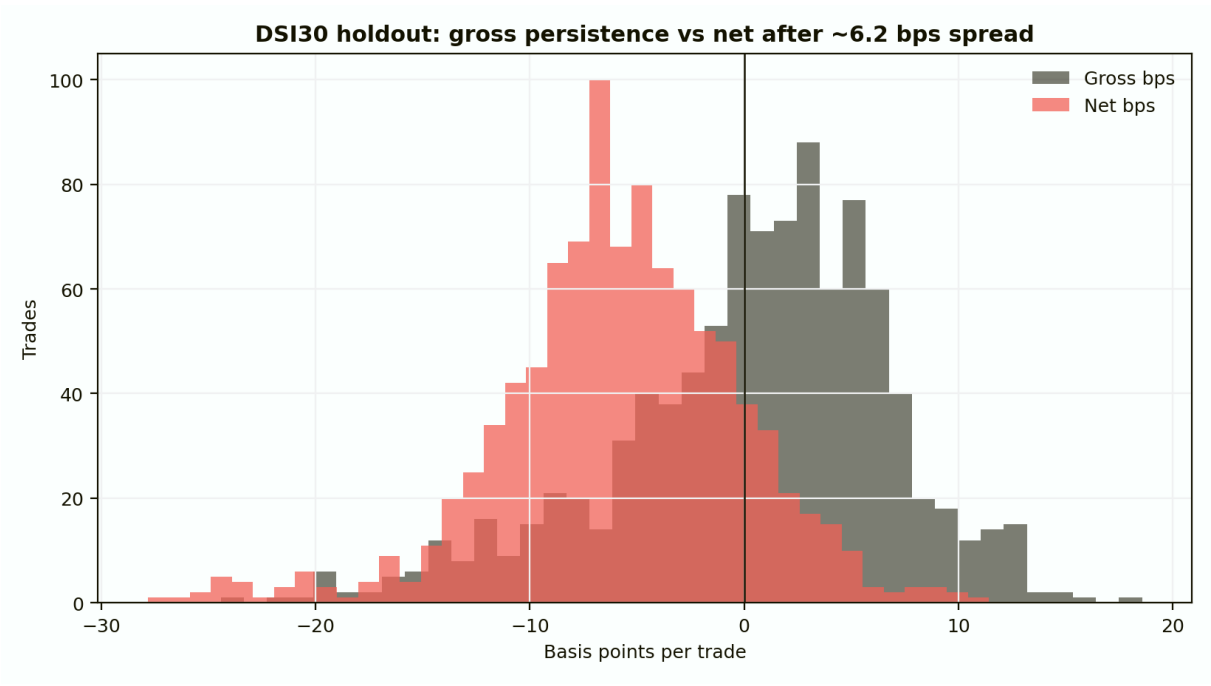


Figure 6. DSI30 holdout gross and net basis points after approximately 6.2 bps spread.

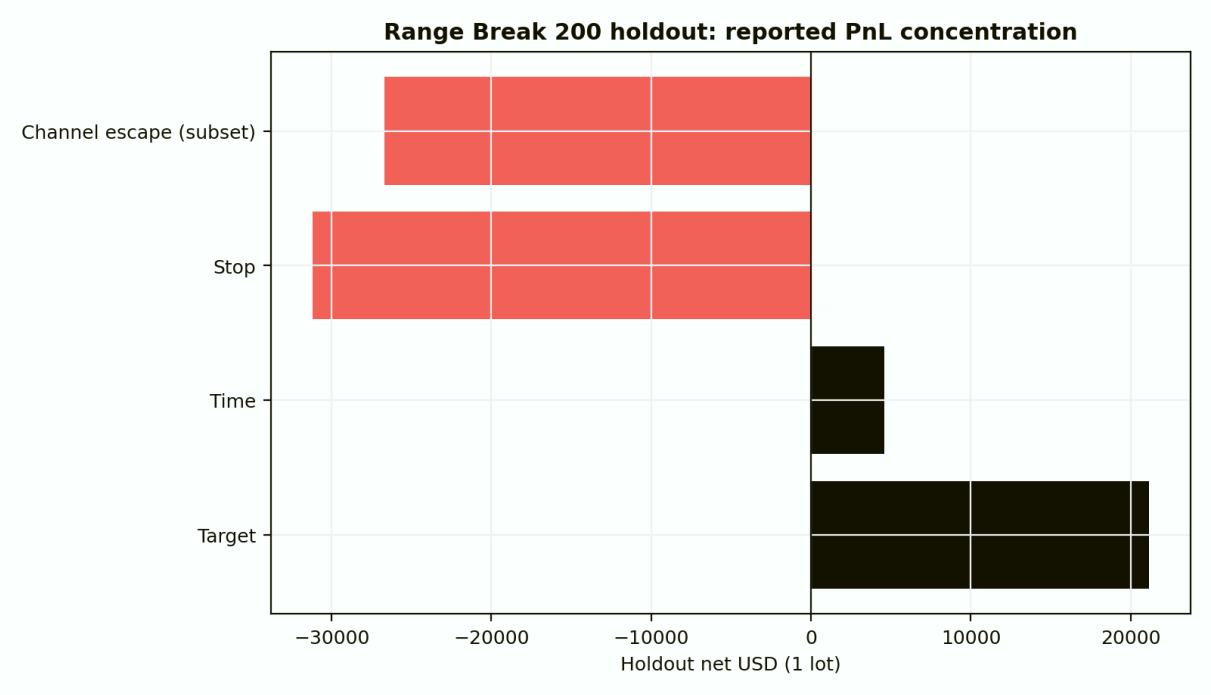


Figure 7. Range Break 200 holdout net USD by exit reason at one lot.

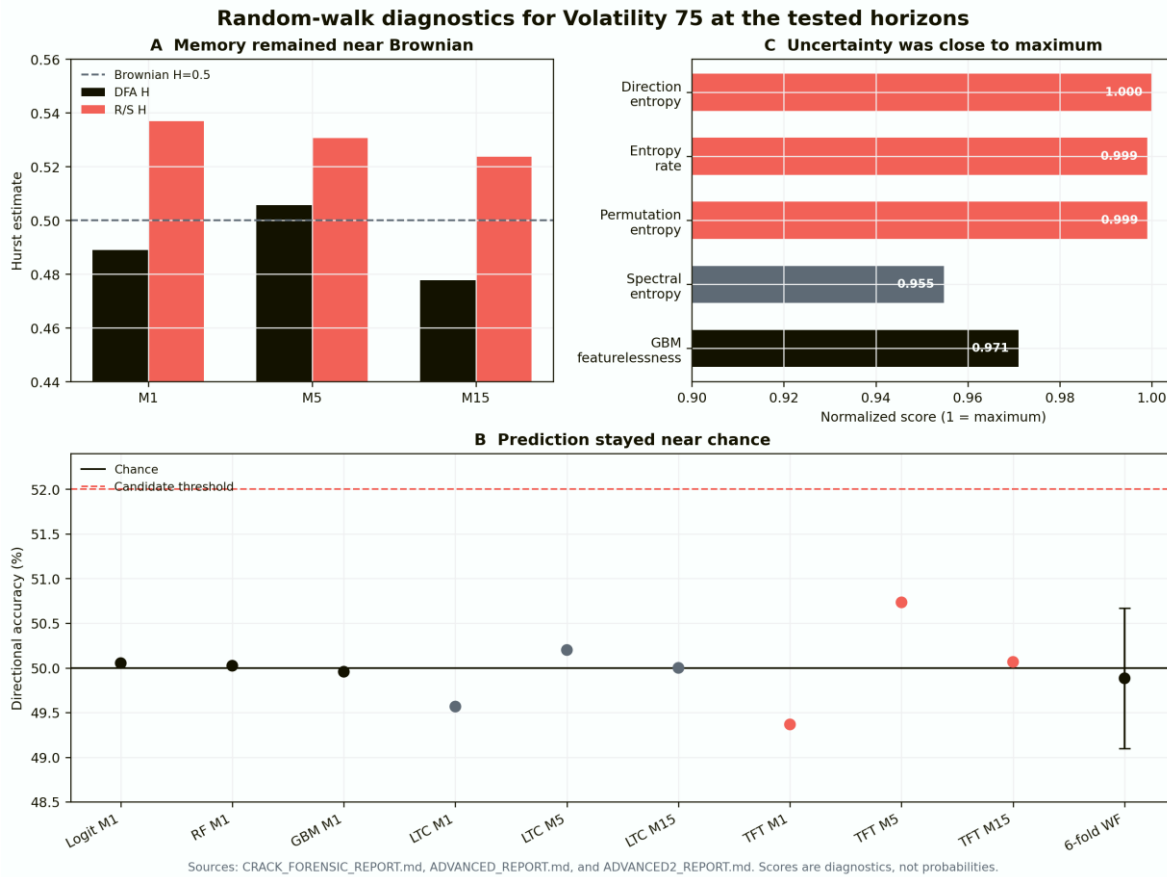


Figure 8. Random-walk diagnostics: near-Brownian memory, near-chance directional prediction, and high normalized uncertainty. Diagnostic scores are not probabilities.